Request for Information; Health Technology Ecosystem
A Notice by the Centers for Medicare & Medicaid Services on 05/16/2025

Source link:
https://www.cms.gov/newsroom/press-releases/cms-seeks-public-input-improving-technology-empower-medicare-beneficiaries

Department of Health and Human Services
Centers for Medicare & Medicaid Services
[CMS-0042-NC]
RIN 0938-AV68

AGENCY:
Centers for Medicare & Medicaid Services (CMS), Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC), Department of Health and Human Services (HHS).

ACTION:
Request for information.

SUMMARY:
Effective and responsible adoption of technology can empower patients to make better decisions for their health and well-being. This request for information (RFI) seeks input from the public regarding the market of digital health products for Medicare beneficiaries as well as the state of data interoperability and broader health technology infrastructure. Responses to this RFI may be used to inform CMS and ASTP/ONC efforts to lead infrastructure progress to cultivate this market, increasing beneficiary access to effective digital capabilities needed to make informed health decisions, and increasing data availability for all stakeholders contributing to health outcomes.

DATES:
To be assured consideration, comments must be received at one of the addresses provided below, by June 16, 2025.

ADDRESSES:
In commenting, refer to file code CMS-0042-NC.

Comments, including mass comment submissions, must be submitted in one of the following three ways (please choose only one of the ways listed):

1. Electronically. You may submit electronic comments on this regulation to https://www.regulations.gov. Follow the “Submit a comment” instructions.

2. By regular mail. You may mail written comments to the following address ONLY: Centers for Medicare & Medicaid Services, Department of Health and Human Services, Attention: CMS-0042-NC, P.O. Box 8013, Baltimore, MD 21244-8013.

Please allow sufficient time for mailed comments to be received before the close of the comment period.

3. By express or overnight mail. You may send written comments to the following address ONLY: Centers for Medicare & Medicaid Services, Department of Health and Human (printed page 21035) Services, Attention: CMS-0042-NC, Mail Stop C4-26-05, 7500 Security Boulevard, Baltimore, MD 21244-1850.

For information on viewing public comments, see the beginning of the SUPPLEMENTARY INFORMATION section.

FOR FURTHER INFORMATION CONTACT:
HealthTechRFI@cms.hhs.gov.

SUPPLEMENTARY INFORMATION:
Inspection of Public Comments: All comments received before the close of the comment period are available for viewing by the public, including any personally identifiable or confidential business information that is included in a comment. We post all comments received before the close of the comment period on the following website as soon as possible after they have been received: https://www.regulations.gov. Follow the search instructions on that website to view public comments. CMS will not post on Regulations.gov public comments that make threats to individuals or institutions or suggest that the commenter will take actions to harm an individual. CMS continues to encourage individuals not to submit duplicative comments. We will post acceptable comments from multiple unique commenters even if the content is identical or nearly identical to other comments.

I. Background
The enactment of the 21st Century Cures Act (Pub. L. 114-255) in 2016 authorized the Centers for Medicare & Medicaid Services (CMS), the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP/ONC),[1] and the National Institute of Standards and Technology (NIST), to implement certain policies to enhance the amount of health data available through digital channels and give patients secure, electronic access to their personal health information. These policies are the building blocks of a digital ecosystem in which patients can view, manage, utilize, and share their health information through digital applications (apps) and other modern solutions. This digital ecosystem could ultimately expand access to personalized health guidance for patients to improve prevention and chronic disease management.

On January 12, 2017, President Trump issued Executive Order 13813, “Promoting Healthcare Choice and Competition Across the United States” (82 FR 48385), which directed Federal

agencies to implement policies that enhance patient access to healthcare data and empower individuals to make informed decisions about their care.

In 2018, CMS led payers by example and released Blue Button 2.0, its own Fast Healthcare Interoperability Resources (FHIR®)-based Patient Access application programming interface (API) with the goal of increasing beneficiary access to their data and improving patient outcomes. This project initiated an ecosystem of patient-facing applications that allowed beneficiaries to authenticate using their Medicare.gov credentials and then to authorize applications to receive their Medicare claims and benefit information. At this time, 75 third-party apps are connected to the Blue Button 2.0 API, giving beneficiaries a variety of services for viewing their health data, sharing their data with digital services, providers, pharmacies, and caregivers, and making decisions related to their healthcare. Currently, Blue Button 2.0 includes Medicare Part A, B, and D claims information, patient demographics, and coverage information. Additionally, in 2019, CMS launched the Data at the Point of Care API pilot to give providers access to claims data for Medicare Fee-For-Service (FFS) beneficiaries they treat.

On May 1, 2020, CMS published the “Medicare and Medicaid Programs; Patient Protection and Affordable Care Act; Interoperability and Patient Access for MA Organization and Medicaid Managed Care Plans, State Medicaid Agencies, CHIP Agencies and CHIP Managed Care Entities, Issuers of Qualified Health Plans on the Federally-Facilitated Exchanges, and Healthcare Providers” final rule (CMS Interoperability and Patient Access final rule) (CMS-9115-F) (85 FR 25510). The rule established standards and requirements for payers regulated by CMS to advance interoperability and data exchange throughout the health system and to signal our commitment to the vision set out in the 21st Century Cures Act and Executive Order 13813 to improve quality and accessibility of information that Americans need to make informed healthcare decisions, including data about healthcare prices and outcomes, while minimizing reporting burdens on affected healthcare providers and payers.

The CMS Interoperability and Patient Access final rule built on the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule, which established the patient's right of access to their protected health information (PHI), including a right to inspect and/or receive a copy of PHI held in designated record sets by covered entities and their business associates as detailed at 45 CFR 164.524. Among other provisions, the 2020 CMS Interoperability and Patient Access final rule required impacted payers to implement and maintain (Health Level Seven (HL7®) FHIR-based APIs that would allow patients to use an app of their choosing to access PHI held by or on behalf of a HIPAA covered entity.

On May 1, 2020, ASTP ONC published the “21st Century Cures Act: Interoperability, Information Blocking, and the ONC Health Information Technology (IT) Certification Program” final rule (ONC Cures Act final rule) (85 FR 25642). This final rule strengthened patients' electronic access (including via a third-party app) to their health information at no cost and added a certification criterion under the ONC Health IT Certification Program to support the availability of FHIR-based APIs in electronic health records and other certified health IT to enable patients and providers to view electronic health information using smartphone applications.

Subsequent ASTP/ONC regulations finalized provisions to advance interoperability, enhance health IT certification, and support the access, exchange, and use of electronic health information. Specifically, the ASTP/ONC “Health Data, Technology, and Interoperability: Certification Program Updates, Algorithm Transparency, and Information Sharing” final rule (HTI-1 final rule), which adopted version 3 of the United States Core Data for Interoperability (USCDI) standard [2] and expanded access to more data via FHIR APIs that meet standards adopted by ASTP/ONC (89 FR 1192); and the “Health Data, Technology, and Interoperability: Trusted Exchange Framework and Common Agreement” final rule (HTI-2 final rule), which implemented provisions related to the Trusted Exchange Framework and Common AgreementTM (TEFCATM) that support information sharing as well as network reliability, privacy, security, and trust (89 FR 101772).

In 2024, CMS published the “Interoperability and Prior Authorization” final rule, which required impacted payers to implement and maintain a Provider Access API, similar to CMS' Data at the Point of Care (printed page 21036) API, to make current patients' claims and encounter data available to in-network or enrolled providers with whom the patient has a verified treatment relationship, at the provider's request (89 FR 8758). The final rule requires impacted payers to be in compliance by January 1, 2027.

The policy framework established by CMS and ASTP/ONC rulemakings are intended to promote the seamless and secure flow of health information between patients, providers, and payers, enabling digital workflows supported by smartphone applications and other modern tools.

II. Solicitation of Public Comments

As the breadth, depth, quality, and timeliness of health data available to patients through standards-based APIs increase, evolving digital health products will gain greater functionality and potential for enhancing the healthcare experience, reducing costs, increasing access to care, enabling chronic disease prevention, and improving healthcare outcomes. Although the building blocks for a patient-centric digital health ecosystem are in place, the experience of most patients, caregivers, and providers is neither seamless nor simple. To get access to their data, patients have to track which providers they have seen and set up separate accounts and credentials with each portal. Even for patients who are able to set this up, digital products for health management or care navigation that can leverage patient health records are still rare; appointment scheduling often still entails lengthy phone calls and provider intake still involves clipboards of multiple forms.

CMS and ASTP/ONC would like to continue to build on the existing policy framework to drive large-scale adoption of health management and care navigation applications, reduce barriers to data access and exchange, realize the potential of recent innovations in healthcare that promote better health outcomes, and accelerate progress towards a patient-centric learning health system.

CMS and ASTP/ONC seek feedback from stakeholders, including but not limited to: patients and caregivers, providers, payers, health IT companies, health information exchanges, health information networks, clearinghouses, researchers, and developers of digital health products regarding how we can help achieve the potential of digital health technology. CMS and ASTP/ONC also seek feedback on which elements of today's digital health ecosystem are working, which are working inconsistently and need improvement, and which are impeding rapid progress. CMS and ASTP/ONC also seek input for possible consideration in future rulemaking on policies to ease health data exchange and promote innovation in consumer digital health products, and how HHS can encourage patient, caregiver, and provider engagement with digital health products.

Furthermore, the transition to value-based care (VBC) represents a cornerstone of CMS' strategy to incentivize improvements in health outcomes rather than increases in service volume. The role of technology is critical to this transformation. While significant progress has been made in health IT adoption, opportunities remain to better align technology requirements with the needs of providers participating in Alternative Payment Models (APMs) and other value-based care programs. Among other topics, CMS and ASTP/ONC seek feedback on current HHS requirements around the use of technology, including the use of health IT certified under the ONC Health IT Certification Program, by healthcare providers participating in VBC initiatives. Among other areas of inquiry, CMS and ASTP/ONC request input on requirements for the use of certified electronic health record (EHR) technology (CEHRT), and how such requirements can enable value-based care and meet statutory requirements while meeting other program objectives, such as reducing provider burden, to better support value-based care adoption among providers, and subsequently improve patient choice and competition in the healthcare marketplace.

The questions that follow are grouped by use cases corresponding to patients and caregivers, providers, payers, technology vendors and data providers, and VBC organizations. We encourage patient advocates and healthcare stakeholders to share their feedback for as many of these use cases as possible. The questions are not meant to be directed to a specific audience but are meant to solicit feedback from multiple types of stakeholders for each use case as appropriate. To aid understanding of submitted responses, please prioritize clarity and conciseness and annotate your responses with question label(s) (for example, PC-1).

Because we expect some responses to relate to software workflows that may be difficult to fully articulate in written responses, we welcome links to screenshots or brief video demonstrations as part of submitted feedback. Please keep videos to a maximum of 15 minutes total to highlight real-world challenges, workflow examples, or functional capabilities and keep introduction of the presenters and company to no more than 2 minutes.

A. Definitions for Terms Used in This RFI

Digital tools: web, mobile or other software applications, potentially leveraging sensors, wearables or other hardware.

Digital health products: defined as digital tools that support individual health needs.
Health management applications: Digital tools that leverage patients' data and other information to support patients with health decision-making.
Care navigation applications: Digital tools that help patients identify, select, and access providers or auxiliary care services.
Personal health record apps: Software applications that collect and organize an individual's health records and provider encounter data for viewing, sharing, and usage in digital health products.

Executive Summary

HealthLX Response to CMS RFI on Health Technology Ecosystem
Reference: CMS-0042-NC | RIN 0938-AV68
Issued by CMS and ASTP/ONC | U.S. Department of Health and Human Services

HealthLX appreciates the opportunity to contribute to CMS and ASTP/ONC's efforts to strengthen the digital health ecosystem through this RFI. We fully endorse the agency's strategic vision to lead with *semantic interoperability* and build upon federal mandates such as the 21st Century Cures Act, CMS-9115-F, CMS-0057-F, and TEFCA. HealthLX brings two decades of data management and large scale application integration experience advancing data integration in complex environments and now offers a proven platform, enabling HL7® and FHIR®—to accelerate compliance, improve care coordination, and enable scalable innovation across Medicare and Medicaid systems.

Driving Semantic Alignment with Composable Automation™

At the core of HealthLX's response is our belief that the healthcare industry cannot achieve scalable interoperability without field-level semantic conformance of the data required by the regulations. We urge CMS to consider an open-source national canonical mapping standard and are prepared to contribute our own canonical model to the ecosystem. As implementers of the CMS regulations for our payer clients since before the inception of the regulations, working as founders of and reference implementers in the Da Vinci project, we've been part of the CMS regulatory journey since the beginning. This experience positions our ideas both as foundational considerations for CMS to consider and secondarily, proof that our work has met the goals of the payers currently on our platform to meet the regulations not just today, but also in the future.

It's for this reason that we believe the inconsistencies in the implementations are directly aligned with the lack of a technical canonical that is foundationally needed to shepard the industry for semantic interoperability and future state compliance objectives.

Our ideas are centered around the implementation philosophy called Composable Automation™ further detailed in the attached document.

Enabling Patient and Caregiver-Centered Experiences

HealthLX aligns with CMS's call to improve the user experience across digital health tools. Our platform can centralize longitudinal patient records across EHRs, payer systems, and third-party applications while incorporating SDOH and behavioral health data. By enabling caregiver access across generations and aligning with initiatives like the Da Vinci Project, the Gravity Project and the Carin Alliance, HealthLX continues to help bridge the equity and usability gaps that hinder effective engagement for vulnerable populations.

Enhancing CMS APIs with Real-Time Intelligence

HealthLX supports CMS's recommendation to expand the regulatory API landscape to include:

- Real-time prior authorization and claims status
- Price transparency and provider directories
- SDOH and pharmacy utilization data

Our platform is purpose-built to ingest, normalize, and orchestrate data from these varied sources in real time. Through our flow editor, monitoring tools, and drag-and-drop configurability, CMS and stakeholders can gain actionable intelligence without high development overhead or prolonged system migrations.

Supporting the Transition to Value-Based Care

The movement to value-based care depends on real-time insight, closed-loop transitions of care, and proactive gap closure. The HealthLX platform is uniquely suited to this challenge—enabling scalable FHIR-native workflows for ACOs, MA plans, and providers. Where most applicable, our integration with legacy and modern API-enabled systems ensures that stakeholders are able to comply with CMS mandates.

Filling Gaps in the Current Ecosystem

Despite the progress made through federal mandates, critical gaps remain in the broader digital health ecosystem:

- Fragmented data architectures and proprietary mapping models
- Limited reuse of integration logic and semantic rules across implementations
- Adoption for the vision of the CMS regulations by all stakeholders including patients
- Poorly implemented solutions for Payers to meet the CMS mandates
- Lag in incorporating SDOH into real-time clinical and coverage workflows

HealthLX addresses these issues through:

- A soon to be announced 'Open-source data mapping canonical' enabling semantic interoperability
- Composable, reusable workflow components to speed to market solutions
- Real-time orchestration of clinical, claims, and community data
- Configurable infrastructure to power required data ingestion at scale issues
- Enabling next-generation patient apps

A Call to Action for CMS

HealthLX respectfully recommends that CMS and ASTP/ONC:

1. Endorse and help govern an open-source, national data mapping canonical model to support both current and future regulations.
2. Expand regulated APIs to encompass additional datasets essential for modern care delivery.
3. Provide implementation incentives tied to semantic conformance and patient engagement metrics.
4. Foster digital equity through caregiver access provisions and usability design requirements.

Commitment to the Future

HealthLX stands ready to support CMS and its partners with:

- Proven infrastructure for HL7®, FHIR®, and complex interoperability mandates
- Real-world deployment experience with Patient Access and Prior Authorization APIs
- A rapidly evolving Composable Automation™ platform that adapts to the needs of value-based care

Together, we can shift from fragmented compliance to seamless, standards-driven innovation—unlocking a truly patient- and caregiver-centered digital health ecosystem.

B. Patients and Caregivers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve patients and caregivers. While we certainly want patients and caregivers to answer questions in this section (and in other sections) from the patient/caregiver point of view, we also invite all stakeholders to provide their viewpoints on the patient/caregiver workflows.

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

A: At HealthLX, we recognize the importance of intuitive and actionable health management tools that empower individuals and caregivers and are philosophically oriented towards patient advocacy.

When we hear the expression 'Provider Burden reduction' we often turn this around to 'Patient Burden Reduction'. Further, while we haven't invested in the direct-to-consumer health management or care navigation apps ourselves, our platform is purpose-built to support and enhance such solutions by enabling seamless data exchange, care coordination, and patient engagement through standards like **HL7 FHIR** for use cases being adopted through **HL7 accelerators (Da Vinci, Gravity and the Carin Alliance).**

This is our near-term mission at HealthLX, to strengthen the data exchange confidence to improve patient care.

The types of health management and care navigation apps we see delivering the most value include:

- **Chronic condition management apps** (e.g., for diabetes, hypertension, or mental health) that track vitals, medication adherence, and lifestyle factors, while providing personalized recommendations.
- **Care coordination platforms** that guide users through post-discharge instructions, referrals, and provider communication—ensuring transitions of care are smooth and well-informed.

- **Digital front doors and symptom checkers**, which help patients navigate urgent vs. routine care and book appointments accordingly.
- **Personal health records (PHRs) and family care apps**, which allow users to manage their own health data as well as that of loved ones, including immunizations, test results, and care plans.
- **Enhanced views of the patients longitudinal record and health journey** will be feasible after the production deployment of the Payer-to-Payer API in January of 2027. Combining views across provider systems and soon multiple payers will enable valuable insights that can guide care.
- **Expansion of FHIR-based data regulations** to include drug related activities and management will further enhance patient engagement and education opportunities.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

A: Digital health products hold tremendous potential to empower individuals and caregivers with more control, insight, and access to healthcare. From our perspective, the top priorities that digital solutions should enable for individuals and their loved ones include:

1. Consolidated Access to Health Data

- View all medical records—from multiple providers, health systems, payers and home health agenies—in one place.
- Seamlessly access claims data, lab results, medications, and immunizations in real time.

2. Personalized Care Navigation & Decision Support

- Receive intelligent recommendations for next steps in care (e.g., book a follow-up, refill a prescription, schedule a preventive screening).
- Get alerts and guidance based on evolving conditions, care plans, or gaps in care.

3. Simplified Communication with Providers & Payers

- Enable secure, direct communication with providers and care teams.
- Enable secure, direct communication to payers care management staffs to better attain guidance of in-network providers to expedite care
- Coordinate care across specialists, primary care, and caregivers—especially critical for chronic or complex conditions.

4. Support for Managing Family & Caregiving Responsibilities

- Manage the health needs of children, aging parents, or others through proxy access and caregiver-specific features.
- Track and manage appointments, medications, and care plans for multiple individuals.
- Incorporate all aspects of care into a centralized application including providers, health systems, payers, pharmacies, home health and even community based organizations.

5. Informed, Cost-Effective Decision-Making

- Compare costs and quality for procedures, providers, and care settings.
- Receive real-time benefit information, prior authorization status, and coverage guidance.

6. Social Determinants of Health (SDOH) Integration

- Identify and address barriers such as transportation, housing, and food access through referral tools and community resource integration.
- At HealthLX we were contracted by HL7/Gravity to build the first reference implementation for FHIR Based workflow for the Gravity project. During this project we established what would be considered universal workflows that the healthcare ecosystem could adopt. The same basic concepts adopted by the Da Vinci project need to be advanced with Gravity by way of standardized APIs for all of the software solutions serving the provider space for these populations.

7. Aggregation of data by way of data exchange does not imply actionable use of the exchanged data.

- One thing about industry recognized standards being adopted by Payers and Providers is the fundamental challenge of accepting the external data as valuable for the member / patient. By default, aggregated patient data however managed and standardized by FHIR / standards bodies will not be useful unless target systems accept this data into their operational systems.
    - *As an example, most providers (and their EHR platforms) do not accept external data as a source of truth. Only additive, if the exchanged data did not originate within the payer or provider. This hinders automation of workflow and higher-confidence engagement with the patient or member. For consumer apps to succeed, with patient control of their data, providers and payers will need to be able to operationalize this data to improve care.*
- Aggregation of all data provides a starting point, but addressing data alignment, data quality, and generation of intelligent summaries for each user role will be required before it is fully actionable. Most users will not dissect a large volume of data to uncover the critical information they may need for a specific patient.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

A: If I had a personal assistant dedicated to supporting my (or my loved ones') health needs, I would prioritize help with the following key areas—many of which are ideally suited for facilitation through emerging and future digital health solutions.

As a reference only, we have been inspired by the life of Mike Morris, who passed away in early 2019 from cancer and founded his company (CureSoft) to solve for care coordination of his own doctors due to their lack of data exchange and sharing. For his survival, he needed to make sure that his care team knew his current state clinical data and created a FHIR-based platform to collect his clinical data from each EHR to provide the insights they needed to care for him in his personal cancer journey.

Mike presented in Washington DC on 12/8/18 to a large audience at an innovation event sponsored by ONC. Here is a link to his presentation for reference.

As we explain the feature needs below that would be added value for our health needs, Mike's story resonates as the broader mission to enable care takers to talk to each other for the survival of the patient.

1. **Coordinating Appointments and Follow-Ups**

- Scheduling primary care visits, specialist consults, labs, screenings, and home health visits—while ensuring they align with care plans and benefit coverage.
- Sending reminders and helping reschedule as needed.

2. **Managing Medications and Refills**

- Tracking current prescriptions, dosage schedules, and potential interactions.
- Coordinating with pharmacies and providers to manage refills or prior authorizations.

3. **Monitoring Health Status and Alerts**

- Tracking vital signs, symptoms, and progress over time, especially for chronic conditions.
- Notifying me (or a caregiver) when values are out of range or action is needed.

4. **Navigating Insurance and Benefits**

- Understanding what services are covered, what costs to expect, and where to go for in-network care.
- Helping with claims status, denials, and prior authorization requirements. (This will be addressed in the Prior Authorization data exchange use case covered in CMS 0057)
- Enabling the members ability to transfer past claims to a new payer should that be required to support any care continuity should insurance coverage transition. (This will be addressed in the Payer to Payer data exchange use case covered in CMS 0057)

5. **Coordinating Care Across Providers**

- Ensuring all members of the care team are aligned on diagnoses, treatment plans, and test results.

- Supporting transitions of care from hospital to home, or between facilities.

6. Accessing and Interpreting Health Records

- Retrieving up-to-date health records across multiple systems.
- Translating clinical jargon into actionable information.
- Auto-generation of personalized health education based on current diagnosis and conditions

7. Supporting Preventive and Proactive Health Goals

- Reminding an individual of vaccines, screenings, and health goals.
- Encouraging healthy habits with nudges and behavior support tools.

At HealthLX, we view this digital assistant concept as very achievable. Through FHIR-based data exchange, configurable workflows, and real-time integration with payers and providers, we enable the orchestration layer necessary to power these types of intelligent, proactive tools.

Whether surfaced through mobile apps, voice assistants, or care platforms, the future of health engagement lies in making this type of virtual assistant experience a reality.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

A: In today's fragmented healthcare ecosystem, centralized access to personal and family health information remains a major challenge. For most individuals—including myself—easy access to a comprehensive view of health data across multiple systems is limited or inconsistent. Records are often siloed within individual provider portals, insurance platforms, or disconnected systems that don't communicate with one another. More specifically, I've experienced the impact of following my primary provider from one network to another here in Wisconsin. Years of my data living in one health system did not follow him into the new health system. This data was digitally shredded, impacting my personal patient journey. I consider this criminal behavior by the provider systems to not share patient data when patients leave a provider network to go to another.

This gap reinforces the critical need for interoperable, FHIR-enabled infrastructure that can consolidate and share health data securely, regardless of where care is received.

b. If not, in what contexts or for what workflows would it be most valuable to (printed page 21037) use one portal or system to access all such health information?

A: Having a single portal or system to access all personal and loved ones' health information would be most valuable in contexts where care coordination, informed decision-making, and timely action are critical. Key workflows where this unified access is especially impactful include:

1. Chronic and Complex Care Management

- Patients managing multiple conditions across various providers need a consolidated view of labs, medications, referrals, and care plans to avoid duplication and conflicting treatments.
- Care teams and caregivers also benefit from synchronized data when adjusting medications or tracking disease progression.

2. Caregiving and Family Health Management

- A caregiver managing appointments, prescriptions, and health decisions for elderly parents, children, or dependents requires permissioned access to their records—ideally in one place, not scattered across provider portals.

3. Emergency and Acute Situations

- In urgent care settings, having one system that surfaces allergies, medications, diagnoses, and recent procedures can dramatically improve care speed and safety—particularly when patients are unable to speak for themselves.

4. Transitions of Care

- Whether moving from hospital to rehab, or primary care to specialty care, seamless access to discharge summaries, referrals, and follow-up instructions is essential to prevent gaps or delays in treatment.

5. **Preventive Care and Health Maintenance**

- Individuals are more likely to follow through on vaccines, screenings, and annual visits when reminded in a centralized system that tracks due dates and gaps in care.

6. **Claims and Coverage Alignment**

- Access to both clinical and claims data in one interface allows users to understand what care they've received, what's covered, and what their next steps should be—bridging the gap between provider and payer information.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

A: Yes, in many cases, certain data types—such as imaging (e.g., x-rays, MRIs), pathology reports, and specialist test results—are either unavailable or inconsistently accessible through existing patient portals or electronic systems. These gaps significantly hinder the ability to make timely, informed decisions about care, particularly when coordinating across multiple providers or care settings.

The key obstacles to accessing and using complete electronic health information include:

1. **Siloed Systems and Lack of Interoperability**

- Health data is often stored in disparate EHRs or departmental systems (e.g., radiology, oncology), many of which don't share data outside their organization or lack standardized formats.
- Even when data exists, it may not be exposed via APIs or shared using FHIR, HL7, or other standardized protocols—preventing aggregation in patient-facing tools.

2. **Limited Functionality in Patient Portals**

- Many portals only display basic information (e.g., appointment history, medications, or lab summaries), with limited ability to view or download full clinical records, images, or notes.
- Imaging results may be referenced but not include the actual images or full radiology reports.

3. Lack of Caregiver or Proxy Access

- Family caregivers often face barriers when trying to access a loved one's records, especially when navigating proxy setup, consent management, or dealing with multiple systems.

4. Gaps in Digital Literacy and Infrastructure

- Patients and caregivers may struggle with portals that are not intuitive or mobile-friendly, or may lack access to reliable internet or devices—particularly in rural or underserved communities.

5. Fragmented Identity and Access Management

- Different health systems require separate logins, making it difficult to consolidate information or maintain a consistent, secure digital identity across platforms.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

A: Yes, several digital health tools are particularly useful for Medicare beneficiaries and their caregivers. These include:

- **PHR & Claims Access Apps** – Platforms like Flexpa, OneRecord and B.Well allow beneficiaries to view, collect and share their Medicare data which originated at the payers, and is accessible through CMS regulation required APIs..
- **Medication Management** – Apps like Medisafe help manage complex medication schedules and reminders.

- **Chronic Condition Support** – Programs such as Omada or Livongo assist with diabetes, hypertension, and lifestyle coaching.
- **Caregiver Tools** – Apps like CareZone provide shared access to health records, calendars, and task lists for family caregivers.

PC-4. What features are missing from apps you use or that you are aware of today?

A: Many health apps on the market today offer valuable features, but there are still several critical gaps that limit their effectiveness—especially for older adults, caregivers, and those managing complex health needs. Key missing features include:

- **Comprehensive Data Integration**
    - Most apps cannot access full clinical and claims data across providers and payers, limiting their usefulness for care coordination and informed decision-making.
    - While our mission at HealthLX is to be an enabling last-mile integration technology, poorly implemented standards and the resulting lack of robust semantic interoperability negatively impact this needed capability.
- **Proxy and Caregiver Access**
    - Few apps support easy, secure management of a loved one's health records, making caregiving tasks harder than necessary.
    - Consent models are requiring government mandates to meet the needed support for proxy and caregiver access. Through the work happening via TEFCA, we remain hopeful to build on these broader consent standards.
- **Contextual Guidance and Actionable Insights**
    - Many tools present raw data (labs, vitals, appointments) without translating it into next steps, health goals, or alerts.
    - Prescriptive guidance with limited provider engagement is the inevitable future of patient care. In order to realize this vision semantic data clarity must be considered the highest of importance in order to progress in this direction.
- **Social Determinants Integration**

    - Limited visibility into non-clinical factors such as transportation needs, food insecurity, or caregiver support—despite their significant impact on outcomes.
    - Movement has occurred around the standards for workflow from the Gravity project, but adoption by providers and payers is limited. Unfortunately, momentum by way of enforcement is needed to occur to solve for the needs of the underserved population.

- **Interoperability and Portability**
    - Most apps are not vendor-agnostic and don't travel well with the patient across systems or life transitions.
    - Because of our belief in data-driven FHIR standards, we acknowledge the differences of interpretation of the CMS guidelines, however the vendors interpreting these data specific field level FHIR resources are inconsistent. To realize the benefits of a universally adopted guidelines by CMS for APIs under regulation, a technically well described canonical model combined with conformance testing needs to be embraced by all healthcare stakeholders..

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

A: There are several types of digital health apps that should exist but are not yet widely available—primarily because of technical, regulatory, or business model barriers. Notable examples include:

1. Truly Integrated Multi-Provider Health Records

- A single app that aggregates clinical, behavioral, and payer data from all sources—primary care, specialists, hospitals, labs, insurance, and community services—without patients needing to log into separate portals.
- Why it doesn’t exist: Persistent interoperability gaps, EHR vendor resistance, and fragmented data ownership make true aggregation difficult.

2. Caregiver-Centric Health Management Platforms

- Apps built primarily for caregivers, allowing them to manage health tasks, view records, coordinate with providers, track insurance activity, and receive alerts across multiple dependents.
- Why it doesn't exist: Most apps are designed for individual users, and proxy access adds legal, privacy, and identity management complexities.

3. Personalized Preventive Health Planners

- An app that integrates age, conditions, risk factors, and benefit coverage to recommend and schedule personalized preventive care—vaccines, screenings, wellness visits, etc.
- Why it doesn't exist: Requires integration with clinical guidelines, payer rules, and scheduling systems—most of which are siloed and inconsistent.

4. Digital SDOH Navigators

- A unified tool that helps individuals find and apply for local support services (e.g., food, transportation, housing) based on eligibility, availability, and location.
- Why it doesn't exist: Requires real-time, localized resource data and partnership models that are still maturing.

Why these gaps remain:

Most of these apps require deep interoperability, seamless consent management, and payer-provider collaboration—infrastructure many systems lack. Additionally, monetization models for apps that prioritize public or caregiver benefit are less developed than direct-to-consumer or health system-sponsored tools.
At HealthLX, we're actively working to close these gaps by enabling the data infrastructure and standards-based exchange needed to support more intelligent, inclusive, and holistic digital health applications based on consistent semantic clarity for FHIR-based data exchange.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

A: As the largest healthcare payer in the U.S., CMS is uniquely positioned to offer workflows that drive system-wide consistency, transparency, and accountability,

especially in areas where alignment across providers, payers, and patients is critical. The most impactful CMS-led workflows would include:

1. **Longitudinal Patient Record Aggregation**
   - CMS can assist in the centralization of clinical and claims data across providers and care settings, giving beneficiaries and their care teams a single source of truth—particularly powerful when paired with Blue Button 2.0 and FHIR access. But enforcement is needed to move the industry forward.
2. **Care Coordination & Transitions of Care**
   - Standardized workflows for discharge planning, care plan updates, and follow-up scheduling across providers and post-acute settings.
3. **Preventive Services & Gap Closure**
   - CMS is well-positioned to proactively notify beneficiaries of recommended preventive services and overdue care based on claims and risk profiles.
4. **Social Determinants of Health (SDOH) Screening and Referral**
   - CMS could lead the standardization of SDOH data capture and referrals to community resources, supporting health equity initiatives and closing non-clinical care gaps.
5. **Authorization and Coverage Transparency**
   - Workflows that allow providers and beneficiaries to understand coverage, prior authorization requirements, and expected out-of-pocket costs in real time.
6. **Population Health and Quality Reporting**
   - Automating performance tracking and value-based care metrics, streamlining reporting for ACOs, Medicare Advantage plans, and other CMS programs.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

A: To encourage greater patient and caregiver interest in digital health products, CMS and its partners should focus on building trust, simplifying access, and demonstrating value by accessing and writing to a data transformation canonical that

all participants can accept, while also enabling versioning of the data mapping canonical with all regulations. Key strategies include:

1. Establish National Standards for Usability and Interoperability

- Promote consistency in digital tools by certifying apps that meet accessibility, FHIR compliance, and usability standards—similar to how CMS certifies EHRs for providers.
- Underwrite and support an data mapping canonical that all parties in the ecosystem can build to.

2. Expand Education and Awareness Campaigns

- Launch public-facing campaigns tailored to Medicare and Medicaid populations, illustrating how digital health tools can simplify care, lower costs, and improve outcomes for individuals and their families.
    - Launch “Did you know…” public service announcements across traditional and social networks to inform the public on their ability to access and utilize their personal data. Short messages sponsored by CMS across a breadth of topics showing the benefit of engaging in your health journey.

3. Incentivize Use Through Benefits and Programs

- Incorporate digital engagement into Medicare Advantage supplemental benefits (e.g., rewards for using medication trackers, wellness apps, or care navigation tools).
- Enable these digital engagement approaches by having a consistent semantic data construct to educate and enable healthcare participants in the interoperable data exchange.

4. Simplify Access and Onboarding

- Integrate digital tools directly into portals patients already use (e.g., MyMedicare.gov) and streamline identity verification and consent processes—especially for caregivers and proxy users.

5. Ensure Equity in Access

- Fund programs that provide devices, training, and internet access to underserved populations, including rural communities and non-English speakers.

6. Leverage Provider Influence

- Encourage providers and care teams to recommend and support digital tools during clinical encounters—patients are far more likely to engage when referred by a trusted professional.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

A: CMS has an important role to play in reviewing and promoting digital health products—not as a regulator of individual apps, but as a standard-setter, data steward, and trust builder across the healthcare ecosystem. Rather than issuing product-specific approvals (as the FDA does), CMS should focus on defining the criteria, data access standards, and programmatic incentives that promote the use of high-quality, impactful digital tools.

a. Role of CMS in Review & Endorsement

CMS should consider a voluntary certification or endorsement framework for digital health tools that:

- Demonstrate measurable impact on care quality, cost, or patient experience.
- Support value-based care models (e.g., ACOs, Chronic Care Management, Medicare Advantage).
- Meet technical standards for FHIR-based interoperability, data privacy, and security.

- Include accessibility and usability benchmarks for seniors, people with disabilities, and caregivers.
- Provide a rapid reporting scorecard to inform the public on application usage, issue classes and privacy breaches.

This type of validation could function similarly to ONC's Health IT Certification program, offering assurance to patients, providers, and payers that tools meet federal guidelines for effectiveness and safety without implying medical endorsement or reimbursement coverage.

b. Recommended Criteria for Review

If CMS were to implement a review framework, suggested criteria should include:

- Adoption to technical specifications that make clear the technical underpinnings of the FHIR resources required for Payer compliance. We believe this should be a data canonical that articulates the specific field level conformance requirements for the APIs under the regulatory mandates.
- Without the above crucial interpretation of the dataflow the following objectives could be envisioned and measured by way of CMS Audits on the Payer and Provider community.
    - Clinical impact and outcomes improvement
    - Evidence of cost savings or efficiency gains
    - Accessibility and equity considerations
    - Interoperability and API compliance (FHIR, OAuth2, etc.)
    - Privacy, security, and consent standards
    - Patient/caregiver usability and adoption rates
    - Support for multi-party access (e.g., proxy, caregiver tools)

c. Policy & Program Levers to Encourage Adoption

To increase adoption of digital health products among patients and caregivers, CMS can pursue:

Enhancements to data access:

- Expand Blue Button 2.0 capabilities, including multi-provider and multi-payer aggregation.
- Simplify identity verification and allow proxy/caregiver delegation via standards-based workflows.
- Reimbursement adjustments: Expand CPT codes for remote patient monitoring (RPM), chronic care management (CCM), and digital coaching.
- Create incentives within Medicare Advantage for using certified digital tools.
- Beneficiary education & engagement:
  - Launch targeted campaigns that explain digital tools in plain language, ideally endorsed by providers and community leaders.
  - Include digital health onboarding in Annual Wellness Visits or discharge instructions.
- Technology ecosystem support:
  - Invest in public/private data utilities that serve as trusted integration layers.
  - Offer open-source frameworks or sandboxes for developers to test FHIR connections and CMS program integration.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

A: To enable timely access to high-quality CMS and provider-generated data, both infrastructure modernization and policy alignment are essential. The following changes would have the most significant impact:

1. Adopt and Enforce Real-Time, FHIR-Based Data Exchange

- Require the technical specifications of the FHIR data construct (i.e., the proposed mapping canonical that HealthLX plans to make available as open-source) and enforce real-time, FHIR-based data exchange. This would enable the data engineering objectives of each health plan to become conformant. If enforced, this would be the baseline for quality measures and a higher level means to audit each payer in their compliance obligations.
- Require broader and deeper adoption of HL7 FHIR APIs for both clinical and claims data, with a focus on real-time or near-real-time updates.

- Extend the existing CMS interoperability rules to cover post-acute, behavioral health, and ancillary care providers, ensuring more complete data flows.

2. Standardize Data Quality and Mapping Requirements

- Mandate consistent coding, terminologies, and value sets (e.g., SNOMED, LOINC, RxNorm, ICD-10) to improve data accuracy and usability across systems. Providers initiate the terminologies, but payers have to agree upon the terminology as part of the adjudication process of a claim.
- Establish data provenance and audit standards so recipients understand the source, context, and reliability of each data element.

3. Strengthen Identity Management and Record Matching

- Implement a unified, patient-centered identity resolution framework, potentially supported by a voluntary national patient identifier or trusted digital ID mechanism.
- Enable federated identity models for secure caregiver and proxy access.

4. Expand CMS Data Access Programs

- Enhance Blue Button 2.0 with support for richer data sets, longitudinal views, and FHIR-based write-back capabilities.
- Offer broader data availability through payer-to-payer APIs, especially during transitions in Medicare Advantage enrollment or dual eligibility.

5. Create Incentives for Provider Data Sharing

- Link reimbursement or quality reporting bonuses to timely data submission into shared health information networks (e.g., TEFCA, QHINs).
- Reduce technical and legal barriers to sharing by expanding data sharing safe harbors and creating plug-and-play integration toolkits with CMS endorsed technical FHIR resources to achieve semantic clarity for data within the FHIR resources.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

A: For Medicare beneficiaries and caregivers—especially those with limited digital experience—simplicity, accessibility, and trust are essential to the successful adoption of digital health products. The most important features include:

1. Intuitive, Senior-Friendly Design

- Clear navigation, large buttons, readable fonts, and minimal steps to complete tasks.
- Avoiding technical jargon; using plain, conversational language.
- Incorporation of personalized health education, or links to trusted sources, based on current diagnosis and conditions.

2. Multi-Modal Access

- Compatibility with smartphones, tablets, desktops, smart TVs, and even voice assistants (e.g., Alexa).
- Offline functionality or printable summaries for those with limited internet access.

3. Guided Onboarding and Support

- Step-by-step tutorials, live chat or call-in support, and caregiver setup guidance.
- Videos or voiceover options for low-literacy or visually impaired users.

4. Caregiver & Proxy Access

- Simple, secure tools for authorized caregivers to access and manage health information on behalf of others.

5. Language and Accessibility Support

- Support for multiple languages, screen readers, closed captions, and compliance with WCAG accessibility standards.

6. Trust Signals and Privacy Clarity

- Transparent privacy policies, clear consent flows, and visible CMS or provider endorsement to build confidence.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

A: If CMS were to collect real-world data on digital health products' impact, the most effective approach would combine **claims-based analysis, clinical data integration, and patient-reported outcomes**, using standards-based infrastructure. Key strategies include:

- **Leverage Existing CMS Data Streams**
  - Use Medicare claims and encounter data to measure cost, utilization, and quality metrics (e.g., readmissions, ED visits, preventive care adherence).

- **Integrate with FHIR APIs and TEFCA Networks**
  - Encourage digital health vendors to expose relevant clinical and engagement data via **FHIR-based APIs**, feeding into CMS registries or data lakes for longitudinal tracking.

- **Include Digital Engagement in Value-Based Programs**
  - Require or incentivize reporting of digital tool usage and outcomes in **ACO REACH, Medicare Advantage, and other value-based models**.

- **Capture Patient-Reported Outcomes (PROs)**
  - Standardize the collection of satisfaction, usability, and health status data directly from users via embedded surveys or portals.

- **Establish a Voluntary Product Registry**
  - Enable vendors to submit de-identified performance data on app usage, health impact, and engagement across populations.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

A: In our experience, the health data that is most readily available and valuable to patients and caregivers includes:

- **Medication Lists and Prescription Histories**
  - Often accessible through EHR portals and pharmacy platforms, these are crucial for managing chronic conditions and avoiding drug interactions.
  - This information is not readily available to care teams responsible for the ongoing requirements to meet the conditions of the patient with chronic care needs.
- **Lab Results and Test Reports**
  - Commonly viewable via provider portals, lab results help patients track trends, understand diagnoses, and prepare for follow-up care.
  - Particularly when not via a referral within a Hospital Care network, this information is not readily available to care teams responsible for the ongoing requirements to meet the conditions of the patient with chronic care needs.
- **Appointment Histories and Future Scheduling**
  - Easily accessible and helpful for tracking care timelines, preparing questions, and managing logistics.
- **Immunization Records**
  - Frequently used for travel, school, or preventive care—especially useful for caregivers of children or elderly adults.
  - A national directory for Immunization records would be helpful to patients that are more mobile, or made available to providers when care is needed. Immunization records are crucial for treatment decisions.
- **Claims and Explanation of Benefits (EOBs)**
  - Available via payer portals; these help patients and caregivers understand service usage, out-of-pocket costs, and coverage.
  - In our opinion, all FHIR resources provided by the CMS guidelines (to include EOBs) should be made available to the patient through 3rd party applications such as Flexpa, B.Well or OneRecord.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

A: Several types of health data are highly valuable but remain difficult to access for patients, caregivers, app developers, and technical vendors due to fragmentation, lack of standardization, or restricted availability. Arguably, this data is available by Payers through provider contracts established with Payers. These include:

- **Claims Data Across Payers**
    - While Medicare claims are accessible via Blue Button 2.0, multi-payer claims data is often siloed, making it hard to get a complete picture of care and costs.
    - Unifying this data to a patient application would present wonderful opportunities to enable better insights into the providers caring for the patients.
- **Clinical Encounter Notes and Operative Reports**
    - Rich in context and insight, but often excluded from patient-facing portals or not shared in structured, FHIR-compatible formats. Transformation of C-CDA documents to FHIR resources is possible currently. HealthLX contributed to the unification of these documents via a FHIR Viewer by way of an ONC sponsored contest in 2016. We took 2nd place in that effort from over 50 entrants. That solution still is operable today, but demand to FHIR enable these documents lacks government endorsed requirements. With CMS enforcement, we believe this data can be made available to patients.
- **Real-Time Appointment Availability and Scheduling Data**
    - Essential for care coordination but inconsistently shared across provider systems, limiting app functionality.
    - In 2021, HealthLX demonstrated a FHIR enabled scheduling prototype via the Da Vinci project at HIMSS. We believe this requirement, if standardized, could also be a feature on 3rd party patient applications

enabling a means to establish appointments across networks while also verifying payer in-network verification to ease patient burden.

- **Accurate, Transparent Pricing Data**
    - Patients and developers still struggle to access actionable cost estimates tied to specific providers, services, and insurance coverage.
    - One of the value adds to 3rd party applications would be comparisons across providers within a geographical area. Should there exist a national database of comparable treatments commonly considered such as 'joint replacements' where services costs could be brought to light, while also verifying payer in-network verification to ease patient burden.

- **Post-Acute and Behavioral Health Data**
    - Often documented in systems that are not connected to mainstream EHR networks or excluded from data-sharing agreements.
    - At HealthLX, we're patient advocates when it comes to privacy about historical treatments that patients may have had in the past. We believe there needs to be consent managed access to this data. This capability does not exist in standards driven by Da Vinci. We believe the ideas evolving in Europe, and particularly in Belgium should be considered to ensure patient consent for all healthcare stakeholders.

- **Social Determinants of Health (SDOH) Data**
    - Valuable for tailoring care and referrals, but under-collected and inconsistently coded.
    - Adoption and enforcement for the use of USCDI SDOH elements by the providers will assist in broader adoption. Again, suggesting adoption to these standards needs enforcement to gain momentum.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

A: Beyond claims and clinical data, several **non-traditional data sources** offer high value for improving health outcomes, care coordination, and patient engagement. The most impactful include:

- **Social Determinants of Health (SDOH) Data**
    - Information on housing, transportation, food security, and social support helps tailor care plans and connect patients to community-based services—especially critical for Medicare and Medicaid populations.
- **Behavioral and Mental Health Data**
    - Often siloed due to privacy rules, yet essential for whole-person care, chronic disease management, and reducing emergency visits.
- **Pharmacy and Medication Adherence Data**
    - Real-time pharmacy fill data and adherence insights support proactive interventions, especially for polypharmacy and chronic conditions.
- **Remote Monitoring and Device Data**
    - Wearables, home health devices, and telehealth platforms generate continuous data that enables early detection, trend analysis, and real-time care adjustments.
- **Home Health and Community Based Organization Data**
    - Senior populations and post-discharge patients are expanding use of home health agencies for nursing, physical therapy, occupational therapy, speech therapy and community based organizations. These providers are often the bridge of information between the patient, their caregiver, and their primary care organization. Capturing critical updates from their patient notes and progress reports would be beneficial.
- **Patient-Reported Outcomes (PROs)**

    - Direct input from patients and caregivers on symptoms, quality of life, and satisfaction offers critical insights into treatment effectiveness and engagement.

- **Caregiver Activity and Coordination Data**

    - Tracking caregiving tasks and engagement helps providers understand support structures and risks for care gaps or burnout.

When integrated alongside claims and clinical data, these sources provide a more complete, personalized view of patient needs—something HealthLX helps enable through interoperability infrastructure and FHIR-based data orchestration.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

A: Opportunities:

- **FHIR and USCDI Expansion:**

    - Broader adoption of **FHIR APIs** and **USCDI standards** across EHRs creates a pathway for structured, queryable data exchange that supports real-time research and decision-making.
    - We continue to be innovators in the standards efforts to support CMS. Our views of the broader needs of the country rely upon data conformance and the related data engineering needs to accomplish conformance.

- **TEFCA Implementation:**

    - The Trusted Exchange Framework and Common Agreement (TEFCA) will help standardize national data sharing across providers, payers, and public health entities.
    - HOWEVER, without a technical data model made available to the country in support of what would be exchanged across TEFCA, adoption will struggle for years. CMS needs to endorse a technical canonical model

that would allow for semantic data sharing objectives to be met.

- **Linking Claims + Clinical + SDOH Data:**
    - Integrating data across domains enables richer, longitudinal datasets for **real-world evidence** generation, predictive modeling, and health equity research.
    - HeatlhLX contributed to the standards work of SDOH sharing by authoring the first SDOH workflow on FHIR by way of the gravity project. The Gravity project continues to evolve based on the original workflow designs created by HealthLX. Continuing support of this project through proper funding will act as a means to enable the mission of SDOH and for the care of the underserved population.

- **Advancements in AI and NLP:**
    - Natural language processing tools can unlock value from unstructured clinical notes, expanding usable data beyond structured fields.
    - While this is a hopeful future state, until data conformance exists, data quality features will not be enforceable since the source systems creating this data will ultimately need to be corrected. Working alongside other open source scoring systems such as Clinical Architectures PIQI frameworks to profile the quality of the data exchange will act as a catalyst for data exchange cleanliness from providers.
    - As a summary point on this data challenge, unless data exchange conformance combined with Data Quality profiling mechanisms are instituted, AI and NLP will not be fully realized. Interestingly, this is the same issue with population health analytics and the insights discovered to lower the cost of care.

Challenges:

- **Data Fragmentation and Siloed Systems:**

- Many providers still operate in non-interoperable environments or use systems that do not fully support standards like FHIR. What is needed by all parties attempting to adhere to the CMS 9115 and 0057 regulations is a canonical model with supporting technical specifications that can be adhered to country-wide. HealthLX has plans to make available an open source canonical with this mission to make the adoption of semantic clarity of the data needed to meet the ecosystem needs.

- **Variable Data Quality and Inconsistent Terminologies:**

  - Differences in coding practices, documentation standards, and data completeness reduce confidence in research-grade datasets.
  - Direction by CMS to enforce a terminology server standardizing CPT and Snomed codes will help clarify expected claim requests from providers and enable more efficient adjudication by payers.

- **Privacy, Consent, and Governance Complexities:**

  - Research use requires clear consent management, de-identification processes, and trust frameworks that many systems are not equipped to handle.
  - Elaboration on privacy and consent should be central to the broader solutions that CMS can encourage. Here is an idea about government sponsored activities in other parts of the world, where the mission is to empower patients with Personal Consent and Data Sovereignty.
  - One of the most pressing challenges in healthcare today is ensuring that patients have meaningful control over their personal health information. As healthcare delivery becomes increasingly digital and interoperable, the U.S. government must prioritize infrastructure that empowers individuals to define and manage consent for the use of their health data.

A promising model for addressing this challenge comes from the work of the Solid Project, created by Sir Tim Berners-Lee and commercialized by Inrupt. This technology enables individuals to store their personal data — including health information — in decentralized “Pods” that they fully control.

External parties such as healthcare providers, researchers, and public health agencies must request access to data, and individuals can grant or revoke that access at any time.

Several international case studies demonstrate how this model can support consent-driven data sharing at scale:

- **Belgium: VITO's "We Are" Platform**
  In Belgium, the Flemish Institute for Technological Research (VITO) launched the "We Are" platform, enabling patients to complete health experience surveys (PREMs) and choose whether or not to share their anonymized results with researchers. The system is built using Solid Pods, allowing for fine-grained personal control over consent (NetCentrix, 2024).
- **Flanders, Belgium: Government-Backed Solid Deployment**
  The Flemish Government is actively deploying Inrupt's Enterprise Solid Server to provide every citizen with a "personal data vault" (Inrupt, 2023). These data vaults allow citizens to manage and share personal data — including health information — with full transparency and consent. Early healthcare use cases include the management of immunization records and health certifications.
- **United Kingdom: NHS Greater Manchester Pilot**
  In partnership with Inrupt, the Greater Manchester region of NHS England piloted a Solid-based health information system aimed at unifying disparate health records under individual control. Patients manage who can access their information across health and care settings (Tech.eu, 2024).
- **Spain: Basque Country Digital Identity and Data Pilots**
  The Basque Country has initiated a government-backed pilot to test Solid Pods for personal data control, with healthcare included as a use case. This approach aims to create a user-centric data ecosystem with secure, consented sharing across public services (Tech.eu, 2024).
- **Australia, Sweden, Argentina**
  Governments in Australia, Sweden, and Argentina have entered contracts with Inrupt to explore Solid-based infrastructure for personal data ownership. These pilots aim to enable consent-driven data sharing in healthcare and other public sectors, aligning with global privacy standards like GDPR (Tech.eu, 2024).

These models illustrate that consent-centric systems can scale nationally and still meet needs for efficiency and interoperability. When individuals trust how their data is

handled, they are more willing to participate in data sharing — benefiting public health, clinical research, and care coordination.

Our Recommendation:

CMS should fund pilot projects and standards development for decentralized personal data stores in healthcare. This includes support for frameworks such as Solid that allow patients to create granular, revocable consent directives, share specific elements of their health data, and ensure interoperability between systems.

By adopting a model that empowers patient control, the U.S. can align with international best practices, restore public trust in data stewardship, and lead innovation in person-centered, consent-driven digital health.

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- **Technical Barriers for Smaller Organizations:**
    - Community hospitals and smaller providers often lack resources to participate in interoperability initiatives, limiting data diversity. Making and supporting a technical mapping standard will enable easier transition for smaller organizations. HealthLX and its vision to provide to the country a canonical for the CMS Patient Access 9115F and Prior Authorization 0057 will be the focus of the supported canonical HealthLX will be open sourcing under an Apache 2.0 licensing agreement.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

A: To make the Blue Button 2.0 API more valuable for digital health developers—and ultimately for patients and caregivers—several additional CMS data sources would significantly enhance its utility:

1. Encounter and Clinical Data (Part C - Medicare Advantage)

- Many beneficiaries are enrolled in MA plans, but Part C data is not currently available via Blue Button 2.0.
- Including this data would provide a more complete view of care utilization and care management activities.

2. Prior Authorizations and Coverage Determinations

- Real-time or historical data on authorization requests and decisions would help patients and caregivers understand service eligibility and navigate care more effectively.

3. Price Transparency and Cost-Sharing Information

- Integrating expected out-of-pocket costs and provider-specific price data would support better financial planning and decision-making.

4. Risk Scores and Quality Metrics

- CMS risk adjustment data and provider quality scores (e.g., Star Ratings, readmission rates) would help apps guide users toward high-quality, efficient care.

5. Social Needs and SDOH-Linked Data

- If available through CMS programs, inclusion of structured SDOH data (e.g., housing instability, food insecurity) could enable more personalized care and service navigation.
- USCDI version 2 incorporated some SDOH data, but adoption to implement is a struggle due to the conformance problem of did the data get populated correctly. HealthLX strongly believes in the resolution of the data conformance problem in order to improve adoption of SDOH visibility to the underserved population.

6. Provider Directory and Network Participation

- Data on in-network providers, including specialty, location, and availability, would make care coordination and appointment scheduling more effective.
- We previously described our desire for FHIR Adoption across payers for scheduling that should be incorporated in patient apps such as Flexpa, OneRecord and B.Well.
- Should data conformance occur and FHIR adoption be universal this information should be enforced by CMS for patient access to their care needs.
- Adding these data types to Blue Button 2.0—through a secure, standards-based API—would enable developers to create **richer, more actionable tools** that support navigation, coordination, and health engagement, particularly for complex Medicare populations.

At HealthLX, we support this vision by enabling **FHIR-based data orchestration and multi-source integration**, helping developers and health plans unlock more value from CMS and payer ecosystems.

a. What difficulties are there in accessing or utilizing these data sources today?

A: Accessing and utilizing additional CMS data sources—beyond basic Blue Button 2.0 claims data—remains challenging due to several systemic and technical barriers:

1. **Limited Availability via Standard APIs**
   Many valuable data types (e.g., Part C data, prior authorizations, provider network participation) are not yet accessible through FHIR-based APIs or self-service developer interfaces.
2. **Fragmentation Across CMS Systems**
   Data is stored in disparate, legacy source databases with different formats, update cycles, and access protocols, complicating integration efforts.
3. **Lack of Real-Time Access**
   Even when data is available (e.g., claims), there is often a significant delay (ranging from weeks to months), limiting its usefulness for timely decision-making and care coordination.
4. **Inconsistent or Incomplete Data Quality**
   Variability in documentation, coding practices, and completeness across

providers undermines data reliability—particularly in areas like social determinants of health and risk scoring. This inconsistency is compounded when data sources from different systems are combined without a shared semantic framework.

5. **Strict Access Controls and Burdensome Approvals**
   Accessing data requires navigating long approval timelines, extensive data use agreements, and demonstrating valid use cases. This is especially burdensome for smaller startups and community organizations.
6. **Patient Identity and Consent Management Gaps**
   Verifying patient identity, managing consent, and enabling secure access for multiple parties (e.g., caregivers) remains a technical and operational challenge.

A Critical Missing Layer: Semantic Clarity Across Stakeholders

These barriers will remain unresolved if the underlying **semantic layer** across stakeholders is not aligned. Without a **mutually agreed canonical data mapping**, discrepancies in how data is defined, coded, and exchanged will persist. This results in confusion, misinterpretation, and reduced data quality, even when technical barriers are addressed.

At HealthLX, we recognize that **semantic consistency** is key to unlocking the true potential of healthcare data. To ensure **FHIR resource semantic clarity**, we are actively considering open-sourcing our **data mapping canonical** to the broader healthcare ecosystem. This would establish a standardized approach to interpreting and exchanging data across stakeholders, ensuring consistent and reliable data integration at scale.

The Need for a Standardized Semantic Data Layer

The open-sourced canonical will offer a standardized framework for stakeholders to map data uniformly, thereby reducing errors, improving interoperability, and enhancing overall data quality.

This effort would complement the work being done with FHIR-based integration layers, enabling secure access, consent workflows, and normalized data across various systems. By adhering to a canonical data layer, we can ensure that all participants—whether they are CMS, healthcare providers, or developers—have a

common understanding of data, enhancing collaboration and improving patient outcomes.

Summary:
These barriers hinder the development of holistic, real-time, and patient-centered tools. At HealthLX, we are addressing these challenges by not only providing FHIR-based integration layers but also leading the effort to establish a **mutually agreed semantic canonical**.

By open-sourcing this mapping, we hope to provide stakeholders in the ecosystem with a clear and unified understanding of healthcare data, ensuring more accurate, interoperable, and efficient data exchanges.
It would be our desire to support this technical data canonical as means to establish semantic clarity in data exchange.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

1. **Expand Available Data Sets**
   To provide a more complete health picture, CMS should include additional data sets such as Medicare Advantage (Part C) encounter data, prior authorizations, price transparency data, and provider directories. By broadening the available data, CMS can empower patients, caregivers, and developers with a more holistic view of healthcare.
   Importance of a Common Canonical: These expanded data sets must adhere to a **mutually agreed-upon canonical data mapping**, ensuring semantic consistency across all new data types. Without this standardized framework, integrating data from diverse sources can lead to inconsistencies, errors, and misinterpretations.

2. **Support FHIR R4 and Future USCDI Versions**
   CMS should transition to **FHIR R4** and align with future versions of the **USCDI** standards. This ensures that Blue Button 2.0 remains compatible with other health data systems, including EHRs and clinical applications, while also providing flexibility to adapt to evolving standards.
   Aligning with a Semantic Canonical: As FHIR continues to evolve, ensuring that both the **FHIR R4** and the **USCDI** implementations adhere to a **mutually agreed semantic canonical** becomes critical for interoperability. This helps to

harmonize FHIR resources with other system standards, minimizing potential conflicts and confusion.

3. **Improve Real-Time Data Availability**
   To enhance care coordination and real-time engagement tools, CMS should work to reduce the latency in claims processing and make more **near-real-time data** available via the Blue Button 2.0 API.
   Real-Time Data and Canonical Alignment: Real-time access to health data requires consistency in how data is mapped and represented. A standardized **canonical framework** can ensure that the real-time data shared via Blue Button 2.0 is both accurate and reliable, helping developers build more responsive and timely applications.

4. **Streamline Developer Onboarding**
   To increase developer participation, CMS should streamline the process of onboarding—simplifying registration, testing, and app review timelines. Offering clear documentation, sandbox environments, and faster approval cycles will ensure faster adoption and innovation in healthcare applications.
   Semantic Clarity for Developers: Along with technical improvements in the onboarding process, providing developers with access to the **canonical data mappings** as part of the documentation can ensure consistency from the start. Clear guidelines on how data should be interpreted and used will reduce integration challenges and enhance the overall developer experience.

5. **Enhance Identity and Consent Management**
   CMS should offer secure, standardized identity verification and flexible proxy/caregiver access options to ensure broader, user-centered engagement. This would allow patients to manage access to their data while ensuring privacy and security.

6. **Canonical Consent Framework**: Managing patient consent is inherently tied to data interpretation. A standardized consent framework—built upon the **canonical data mappings**—would ensure that consented data is shared consistently, maintaining both security and clarity. This approach minimizes errors and increases patient trust by ensuring that all parties interpret consent in the same way.

7. **Provide Clear Data Provenance and Metadata**
   To help developers assess the reliability and freshness of the data, CMS should include metadata such as **timestamps**, **source system indicators**, and **versioning**. This would allow developers to assess data accuracy and decide how it should be used in their applications.

8. **Data Provenance and Canonical Consistency**: In addition to metadata, adhering to a **mutually agreed canonical** for data mapping ensures that data provenance is clear and consistent. This canonical framework would include rules for versioning and data updates, ensuring that developers can trace the history of the data with confidence and rely on its accuracy when integrating it into applications.

c. Is there non-CMS data that should be included in the API?

A: Yes, integrating select non-CMS data sources into or alongside the Blue Button 2.0 API would provide a more complete, actionable view of a patient's health, especially for care coordination, chronic disease management, and caregiver support. Key data sources include:

1. Provider EHR Data

- Clinical notes, test results, care plans, and encounter details from non-Medicare providers would give users a more holistic clinical view, especially for dual eligibles or those with supplemental coverage.

2. Pharmacy and Medication Adherence Data

- Real-time fill status and adherence metrics from retail and mail-order pharmacies support safer medication management and improved outcomes.

3. Behavioral Health and Post-Acute Care Data

- Often siloed from CMS systems, these records are critical for comprehensive care planning and monitoring.

4. Remote Patient Monitoring (RPM) and Wearable Device Data

- Home health and biometric data provide ongoing insight into daily health trends and allow earlier interventions.

5. Social Determinants of Health (SDOH) and Community Resource Data

- Including data from community-based organizations or referral platforms (e.g., Unite Us, findhelp.org) would support care plans that address non-clinical needs.

Including these sources—either through linked APIs or supported integration frameworks—would dramatically increase the value, context, and personalization of digital health tools connected to Blue Button 2.0.

PC-10. How is the Trusted Exchange Framework and Common AgreementTM (TEFCATM) currently helping to advance patient access to health information in the real world?

A: The **Trusted Exchange Framework and Common Agreement (TEFCA)** is laying critical groundwork to improve **patient access to health information**, though its full impact is still emerging as implementation progresses.

How TEFCA is helping today:

1. **Standardizing Nationwide Data Exchange**

    - TEFCA creates a **unified, trust-based framework** that allows health information networks (QHINs) to exchange data seamlessly—removing barriers between systems that previously did not interoperate.

2. **Building Public Infrastructure for Patient-Centered Access**

    - TEFCA includes provisions for **individual access services**, enabling patients to retrieve their data from participating QHINs through approved third-party apps or services, regardless of where they received care.

3. **Promoting Equity and Portability**

- By requiring consistent technical and policy standards across networks, TEFCA supports **equitable access** to data for patients in rural or underserved areas who rely on smaller or less-connected providers.

4. **Laying the Foundation for Scalable Interoperability**

    - As more stakeholders come on board to TEFCA, the potential for **real-time, longitudinal data access** across payers, providers, and care settings increases—empowering patients with a fuller picture of their health.

At HealthLX, we are building solutions that align with TEFCA's goals—supporting **FHIR-based connectivity, consent workflows, and multi-party access models** that will enable patients and caregivers to benefit fully from TEFCA-enabled data exchange.

a. Please provide specific examples.

Though still early in rollout, TEFCA has begun enabling patient access in specific ways:

- **QHIN Designation**: Early Qualified Health Information Networks (QHINs) like **CommonWell Health Alliance**, **eHealth Exchange**, and **Epic's TEFCA connection** are actively onboarding participants, allowing data to move across previously siloed networks.

- **Individual Access Services (IAS)**: TEFCA requires that QHINs support **third-party apps** to retrieve patient data on request. For example, a patient using a health app may soon be able to access their longitudinal record across different hospitals and clinics participating in TEFCA, regardless of EHR vendor.

- **Provider-to-Provider Exchange**: As QHINs begin exchanging data, patients benefit indirectly through **better care coordination**, fewer redundant tests, and faster record transfers when moving between health systems.

These early steps demonstrate TEFCA's potential to enable **seamless, patient-authorized access** to comprehensive health records—regardless of geography or provider network.

b. What changes would you suggest?

A: To accelerate TEFCA's impact on real-world patient access, we recommend the following changes:

1. **Fast-track Individual Access Services (IAS) Implementation**
    - Clearly define and prioritize timelines for when and how patients can use third-party apps to access their data through QHINs.
2. **Support FHIR-Based Data Exchange**
    - Expand TEFCA's exchange capabilities to include **FHIR APIs**, not just document-based exchange (e.g., CCDAs), to enable real-time, app-friendly data access.
    - FHIR-Based data exchange is the inevitable target technical means for data exchange. This shouldn't be an 'option' it should be the mandate for all TEFCA exchanged documents with the supporting technical understanding and testing mechanisms necessary to validate the data assuring source system data is providing acceptable semantic interpretation of the payloads being exchanged.
3. **Enhance Patient and Caregiver Education**
    - Launch CMS-led outreach explaining how TEFCA benefits patients and caregivers, and how to access data through approved tools.
4. **Simplify Identity and Consent Management**
    - Provide a **federated identity model** and consistent standards for proxy and caregiver access across QHINs.
5. **Expand Participation to Payers and Community Providers**
    - Include more **Medicare Advantage plans, Medicaid programs, and behavioral health providers** to ensure complete longitudinal data.

These changes would help TEFCA deliver on its promise of **trusted, nationwide, patient-centered data access**

c. What use cases could have a significant impact if implemented through TEFCA?

A: Several **high-impact use cases** could be significantly advanced through TEFCA implementation:

1. **Longitudinal Health Record Access**
   - Empower patients and caregivers with a **comprehensive view of their health history** across multiple providers and care settings.
2. **Transitions of Care**
   - Seamless data exchange between hospitals, primary care, specialists, and post-acute providers can reduce **readmissions and care delays**.
3. **Emergency Care Access**
   - Real-time access to key clinical data (e.g., allergies, meds, conditions) can improve **decision-making in emergency settings** when a patient can't speak for themselves.
4. **Caregiver and Proxy Data Access**
   - TEFCA can enable secure, permissioned access for **family caregivers managing care** for elderly or disabled loved ones.
5. **Population Health and Risk Stratification**
   - Aggregated data across QHINs supports **better targeting of preventive care**, chronic condition management, and value-based interventions.
6. **Social Services Coordination**
   - Linking healthcare data with **social determinants of health (SDOH)** and community-based services enables more holistic care planning.

By supporting these use cases, TEFCA can help drive **better outcomes, equity, and patient engagement** at scale.

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

A: Several **interoperability standards** are currently working well to advance access and support real-world data exchange:

1. HL7® FHIR® (Fast Healthcare Interoperability Resources)

- The foundation for modern, app-based access to clinical data.
- Widely adopted for **patient access APIs**, **payer-to-payer data exchange**, and **provider integration**.

2. USCDI (United States Core Data for Interoperability)

- Defines the **minimum data set** required for standardized exchange, ensuring consistency in data shared across systems and programs.

3. OAuth 2.0 and SMART on FHIR

- Enable **secure authentication and authorization** for patient-directed data access through third-party apps, aligned with CMS and ONC rules.

4. HL7® C-CDA (Consolidated Clinical Document Architecture)

- Still commonly used in provider-to-provider exchange and supported in TEFCA's current framework, especially for summary of care records.

5. X12 for Claims and EDI Transactions

- While more legacy in nature, X12 remains reliable for **payer-provider exchanges** related to claims, eligibility, and coverage.
- **X12 data constructs,** while reliable, remain a bottleneck for innovative ways to interact in a real-time API economy. At HealthLX, we prototyped various X12 data constructs in FHIR, which is increasingly used around the world in lieu of X12 for claims, eligibility and coverage.

These standards form the backbone of many CMS interoperability initiatives and TEFCA-aligned networks. At HealthLX, we actively support integration across these

standards to help clients enable **secure, real-time, and patient-centered data exchange**.

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

A: Several promising standards are not yet widely implemented but could significantly improve data access, integration, and patient-centered exchange:

1. HL7® FHIR® Bulk Data (Flat FHIR / Flat File Export - $export)

- Enables **population-level data exchange**—ideal for research, quality measurement, and value-based care programs.
- Underutilized due to infrastructure limitations and lack of mandates outside select CMS initiatives.

2. Gravity Project SDOH FHIR Implementation Guides

- Standardizes **collection and exchange of social determinants of health (SDOH)** data.
- Crucial for holistic care, but adoption is still limited across EHRs and community organizations.

3. FHIR Write Capabilities (Read/Write APIs)

- Most current APIs are read-only. Enabling **secure data write-back** would support care plan updates, appointment scheduling, and digital intake—transforming patient and app engagement.

4. Consent and Authorization Standards (e.g., HL7 FHIR Consent Resource)

- Enables standardized, digital **consent management** for patients and proxies across organizations—a key enabler for broader access.

5. Digital Identity Standards (e.g., OpenID Connect, UDAP)

- Critical for **trusted identity verification**, especially across networks and when enabling individual or caregiver access through apps.

Greater adoption and alignment around these standards would unlock **more scalable, secure, and meaningful data exchange**—especially in support of TEFCA, CMS interoperability mandates, and patient-directed access.

f. Are there redundant standards, protocols, or channels that should be consolidated?

A: Yes, several redundant standards and exchange mechanisms could be consolidated to reduce complexity and improve interoperability.

For example, C-CDA and FHIR are often used for similar clinical data exchange purposes; consolidating around FHIR would support more modular, modern, and developer-friendly workflows. Likewise, fragmentation in identity and consent frameworks (e.g., OAuth, HL7 Consent) could be addressed by establishing a unified, patient-centric model.

Data exchange channels—such as Direct messaging, X12 transactions, and FHIR APIs—often overlap in functionality, and transitioning toward FHIR-based APIs would streamline infrastructure across payers and providers.

Additionally, harmonizing overlapping reporting standards (e.g., MIPS, HEDIS, eCQMs) using FHIR-based formats like MeasureReport would reduce burden and enable more meaningful use of reported data. HealthLX supports this transition by providing a flexible, FHIR-native interoperability platform that unifies disparate data standards and exchange workflows—accelerating the shift to scalable, standards-driven infrastructure.

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

A: While there are existing alternatives outside of TEFCA that support patient access to health information, they are fragmented, inconsistent, and not yet adequate for achieving truly widespread, seamless access.
Key alternatives include:

- **FHIR-based Patient Access APIs (mandated by CMS and ONC)**
    - These allow patients to connect apps to retrieve data from individual payers or providers. However, access is limited to single systems,

requiring multiple logins and offering incomplete records.

- **Blue Button 2.0**
    - Provides Medicare beneficiaries access to their claims data, but excludes Part C (Medicare Advantage) and has limited clinical depth.
- **Third-Party Health Apps (e.g., Apple Health, CommonHealth)**
    - These can aggregate data from participating providers but depend heavily on EHR vendor and provider participation, limiting reach.
- **Health Information Exchanges (HIEs)**
    - Some regional or state-level HIEs provide portals for patients, but participation is inconsistent and not national in scope.

Despite these options, there is no unified, scalable mechanism for national, cross-system patient access—especially for individuals managing care across multiple providers and payers. TEFCA is uniquely positioned to close this gap by creating a trusted, standardized, and interoperable framework for broad, secure access across the healthcare ecosystem.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

A: Health Information Exchanges (HIEs) play a pivotal role in enhancing patient access to health information by facilitating secure, electronic sharing of medical data among healthcare providers. This connectivity ensures that patients and their care teams have timely access to comprehensive health records, which is essential for coordinated and informed care.

Key Contributions of HIEs to Patient Access:

- **Consolidated Health Records:** HIEs aggregate patient data from various sources—such as hospitals, clinics, laboratories, and pharmacies—into a unified record. This comprehensive view enables patients to access their complete medical history, including medications, allergies, test results, and treatment plans, through patient portals or upon request.

- **Improved Care Coordination:** By ensuring that all members of a patient's care team have access to the same up-to-date information, HIEs reduce redundant tests and procedures, minimize medication errors, and enhance overall care quality.
- **Emergency Access:** In urgent situations, authorized providers can quickly retrieve critical patient information via HIEs, facilitating prompt and accurate treatment decisions even if the patient is unable to communicate.
- **Empowering Patients:** Some HIEs offer consumer-mediated exchange capabilities, allowing patients to actively manage and share their health information with chosen providers, fostering greater engagement in their own care.
- **Visibility to Fraud:** Patients 'working the system' will attempt treatment at multiple provider facilities to ensure access to medications they have been subscribed to. HIEs provide insights into these situations that otherwise would go unnoticed.

Real-World Example:
The Mount Sinai Health Information Exchange in New York exemplifies the benefits of HIEs. It enables authorized healthcare professionals to access a wide range of patient information—such as allergies, medications, and lab results—across participating facilities, thereby enhancing the continuity and quality of care. Mount Sinai Health System

In summary, HIEs significantly advance patient access to health information by creating interconnected systems that support comprehensive, timely, and patient-centered care while enabling cross system analytics enabling insight into Fraud or misuse of provider services.

a. How valuable, available, and accurate do you find the data they share to be?

A: The data shared through Health Information Exchanges (HIEs) is often highly valuable, particularly for ensuring continuity of care across providers and settings. HIEs help deliver critical clinical information—such as medications, lab results, allergies, and encounter summaries—that patients and their care teams may not otherwise have access to in real time.

That said, data availability and accuracy vary widely depending on the participating organizations, level of integration, and local infrastructure:

- Availability can be limited if providers do not contribute data consistently or if certain systems (e.g., behavioral health, post-acute care) remain disconnected.

- Accuracy issues can arise from duplicate records, inconsistent coding, delayed updates, or incomplete patient matching.

An interoperability platform is needed to address these challenges by providing a FHIR-native interoperability platform that enhances the value of HIE data. As an example, here are considerations for the needed interoperability platform that should be adopted:

- Normalize and enrich data received from HIEs and other sources to improve consistency and usability.

- Bridge gaps in HIE coverage by integrating additional payer and provider data sources.

- Support identity resolution and consent workflows to ensure that data is securely shared and accurately attributed to the right individual.

b. What changes would you suggest?

A: To maximize the value of HIEs for patient access and care coordination, several improvements are needed—especially in alignment with HealthLX's interoperability approach:

- Adopt FHIR-Based APIs Across HIEs
- Transition from legacy document exchange (e.g., C-CDA) to FHIR APIs for more granular, real-time access to clinical data.
- Standardize Data Quality and Normalization Practices
- Implement consistent terminology mapping (e.g., SNOMED, LOINC) and data validation frameworks to improve the accuracy and usability of shared data.
- Enhance Patient Identity Matching and Consent Management

- Support stronger identity resolution protocols and standardized proxy/caregiver consent workflows to reduce errors and ensure secure, authorized access.
- Expand Participation and Data Types
- Encourage participation from behavioral health, post-acute, and social service providers to provide a more complete view of patient care.
- Integrate with National Frameworks Like TEFCA
- Align local and regional HIE efforts with TEFCA to support nationwide, trusted exchange and reduce fragmentation across networks.

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

A: Yes, Here are some examples:

1. CRISP (Chesapeake Regional Information System for Our Patients – Maryland/DC region)

- Model Strengths: Offers real-time notifications, comprehensive patient summaries, and integrated services for public health, payers, and providers.
- Patient Impact: Enables authorized access to critical health data across care settings, including behavioral and social services.

2. Manifest MedEx (California)

- Model Strengths: Aggregates data from over 120 hospitals, 400 provider organizations, and major payers to support care coordination, population health, and analytics.
- Patient Impact: Facilitates longitudinal views and supports value-based care programs, including COVID-19 public health efforts.

3. Indiana Health Information Exchange (IHIE)

- Model Strengths: Long-standing infrastructure with strong data governance and statewide provider engagement. Offers predictive analytics and care quality support.
- Patient Impact: Improves care transitions and reduces duplication through robust data sharing and alerting tools.

4. MIHIN (Michigan) - Strengths and Patient Impact

Strengths

- **Extensive Data Aggregation:** Connects over 120 hospitals, 400 provider organizations, and major payers across Michigan.

- **Infrastructure for Coordination:** Enables robust **care coordination** and **population health management** through real-time and historical data exchange.

- **Analytics-Ready Data:** Supports **healthcare analytics** initiatives with a standardized data backbone, enhancing insights at both patient and population levels.

Patient Impact

- **Longitudinal Health Records:** Gives providers a **comprehensive view of patient history**, improving continuity of care—especially for patients seeing multiple providers.

- **Support for Value-Based Care:** Empowers providers to **track outcomes and close care gaps**, enhancing the effectiveness of **preventive and chronic care** management.

- **Public Health Readiness:** Played a critical role during COVID-19, **accelerating data exchange** and coordination for **testing, treatment, and vaccination** tracking.

These HIEs succeed because they go beyond basic data exchange, offering integrated services, analytics, and real-time alerts that enhance care delivery and empower patient access.

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

A: HIEs continue to play a complementary role in the evolving health data exchange landscape—particularly in supporting localized, real-time, and community-specific data sharing. While frameworks like TEFCA and national vendor networks (e.g., CommonWell, Carequality) enable broad interoperability, HIEs offer value in several distinct ways:

- **Local Care Coordination Infrastructure**
    - HIEs maintain close ties with regional providers, public health agencies, and community organizations, enabling context-aware, real-time exchange tailored to local needs.
- **Integration of Non-Traditional Data Sources**
    - Many HIEs are uniquely positioned to incorporate behavioral health, post-acute care, social services, and public health data that are often excluded from national networks.
- **Public Health and Emergency Response Support**
    - HIEs have proven essential in disease surveillance, contact tracing, and vaccination tracking, particularly during COVID-19 and regional emergencies.
- **Intermediary Between Local and National Exchange**
    - HIEs can act as trusted nodes or gateways within TEFCA and private frameworks, bridging data between local networks and national infrastructure.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

a. Examples may include the following:

(1) Binding cost estimates for pre-defined periods.

Patients and caregivers need reliable, upfront knowledge of healthcare costs. Binding cost estimates—based on a patient's plan, provider, and service bundle—would reduce financial uncertainty, increase confidence in care choices, and enable comparison shopping.

(2) Viewing provider schedule availability.

Access to real-time provider schedules would streamline care access, reduce delays, and promote provider competition based on convenience and responsiveness.

(3) Using third-party apps for appointment management.

Allowing patients to schedule, cancel, and modify appointments via trusted third-party applications promotes interoperability and reduces friction—especially for patients seeing multiple providers.

(4) Accessing patient-facing quality metrics.

Patients benefit from transparent quality measures such as outcomes, safety ratings, and patient experience scores. When personalized and contextualized, these metrics can guide more informed provider selection.

(5) Finding the right provider for specific healthcare needs.

Patients need tools that combine availability, cost, network status, specialty, quality, and patient preferences. Integrating these factors into a single, intelligent search interface would improve care matching and reduce disparities in access.

Conclusion:
These operational use cases are central to a more transparent, competitive, and patient-centered health system. HealthLX supports these efforts by providing a configurable, scalable platform that enables secure, standards-based integration of clinical, claims, scheduling, and quality data—ensuring patients and caregivers have the information they need, when and where they need it.

b. What use cases are possible today?

While many operational data use cases are still maturing, several are technically possible today, especially with support from CMS regulations, payer mandates, and emerging interoperability standards. However, current implementation is often fragmented and inconsistent across systems. The following use cases are feasible now, and HealthLX supports their deployment through FHIR-based integration and workflow orchestration:

1. Viewing Provider Directory Information

- Patients can access provider names, specialties, locations, and network status through CMS-mandated payer APIs. However, this data often lacks real-time availability or scheduling information.

2. Accessing Basic Cost Transparency Tools

- Some health plans and provider groups offer cost estimation tools based on historical claims and benefit design, though these are often limited to a subset of services and lack precision.

3. Third-Party Appointment Integration (Limited)

- Certain EHRs and digital health tools enable appointment scheduling or reminders through APIs, but access is often vendor-specific and not standardized across providers.

4. Access to Public Quality Metrics

- Patients can find hospital and provider quality scores through public CMS datasets (e.g., Hospital Compare, Star Ratings), but the data is often static and not personalized.

5. Manual Care Navigation Tools

- While not always automated, some apps offer condition-based provider search, using filters such as specialty and proximity. These tools often rely on static data sources or require users to piece together fragmented information.

Conclusion:
These use cases represent what is possible with today's data standards and APIs—but full realization of their value depends on deeper integration, improved data quality, and support for real-time workflows.

c. What should be possible in the near future?

In the near future, patients and caregivers should be able to access a fully integrated, user-friendly digital ecosystem that combines real-time operational, clinical, and financial data to support better care navigation, reduce administrative burden, and increase market competition. With advancing interoperability standards and national frameworks like TEFCA, the following use cases should be achievable at scale:

1. Real-Time Provider Schedule Availability and Booking

- Patients should be able to view live appointment slots across providers and book care directly through third-party apps, regardless of EHR or practice management system.

2. Personalized, Coverage-Aware Cost Estimation

- Patients should be able to see out-of-pocket estimates based on their benefits, provider network status, and service specifics—before scheduling or receiving care.

3. Integrated Provider Search with Quality + Access + Cost Data

- Patients should be able to search for providers not just by specialty or location, but also by availability, cost efficiency, outcome measures, and patient satisfaction.

4. Multi-Provider Appointment Management Tools

- Patients managing care for multiple conditions or family members should be able to coordinate appointments across systems, with reminders and care plan alignment built in.

5. Dynamic Referrals and Smart Care Pathways

- Referrals should be based on availability, specialization, quality, and cost—automatically routed and tracked across systems.

Conclusion:
These future-forward use cases are increasingly feasible due to maturing standards (FHIR R4+, USCDI v3), TEFCA infrastructure, and CMS API mandates. HealthLX stands ready to support their implementation through standards-compliant, extensible, and payer-provider integrated interoperability solutions.

d. What would be very valuable but may be very hard to achieve?

Several operational health data use cases offer **tremendous value for patients and caregivers**, but remain difficult to implement due to entrenched silos, limited incentives, or technical and regulatory complexity. These challenges can be overcome with the right infrastructure, policy alignment, and partner collaboration—but they represent long-term goals.

1. Real-Time, Binding Cost Estimates Across Complex, Multi-Provider Episodes

- Patients should be able to receive **binding, personalized cost estimates** for bundled services (e.g., joint replacement, cancer treatment) that include all related providers and facilities.
  Barrier: Requires coordination across billing systems, negotiated rates, benefit details, and authorization logic.

2. Fully Interoperable Appointment Booking Across All Providers and Platforms

- Patients should be able to book appointments with any provider—regardless of system, specialty, or network—through trusted digital tools.
  Barrier: Fragmentation in scheduling systems, lack of standardized APIs, and limited EHR vendor cooperation.

3. Integrated Provider Selection Engine With SDOH, Risk, and Outcomes Insights

- Patients should be guided to the “best fit” provider using real-time data on outcomes, patient demographics, risk-adjusted performance, and social needs.
  Barrier: Requires standardized, accessible outcomes and SDOH data, plus decision support logic that integrates with patient preferences.

4. Cross-System, Longitudinal Appointment and Treatment Plan Coordination

- Caregivers should be able to manage care for a loved one across multiple specialists and facilities with a unified view of schedules, care plans, and required follow-ups.
  Barrier: Proxy access, data permissions, and workflow alignment across systems remain inconsistent.

5. Automated, Patient-Facing Prior Authorization with Timeframes and Outcomes

- Patients should be able to initiate, track, and receive updates on **prior authorization** requests in real time, with clear turnaround times and alternatives if denied.
  Barrier: Proprietary payer workflows, lack of standardization in prior auth data

exchange, and absence of patient-facing status visibility.

Conclusion:
While these use cases are technically achievable, their complexity lies in cross-entity coordination, policy modernization, and standardization enforcement. HealthLX is committed to advancing these goals by delivering **interoperability infrastructure that simplifies integration and scales securely across ecosystems**.

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

A: CMS can encourage patients and caregivers to report information blocking by embedding clear prompts and educational messages in tools like MyMedicare.gov, EOBs, and enrollment materials, and by partnering with providers and app developers to raise awareness. Increased reporting would spotlight real barriers to interoperability, incentivize compliance, and help build patient trust in digital access.

While a surge in complaints may initially reveal systemic gaps, the long-term impact would be positive—advancing data exchange, informing policy improvements, and reinforcing patient rights. HealthLX supports this effort by providing FHIR-based infrastructure that minimizes the technical causes of information blocking.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

Patients and caregivers face several challenges in adopting digital identity credentials, including complex enrollment processes, low awareness, and limited interoperability across systems. Many users—especially seniors or those with low digital literacy—struggle with document verification or biometric steps, while others are hesitant due to privacy concerns.

Additionally, underserved populations may lack the necessary ID or internet access to complete the process. To improve adoption, CMS could simplify credentialing

workflows, endorse trusted providers, and promote inclusive, accessible digital identity education. HealthLX supports integration with credentialing services to enable secure, user-friendly access to health data.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

Widespread use of digital identity credentials would offer significant benefits to patients and caregivers, including **simplified and secure access** to health records across multiple systems, apps, and providers. With a single, verified identity, users could more easily manage their care, authorize third-party tools, and coordinate across providers without juggling multiple logins or paperwork.

For caregivers, standardized digital identity would support **proxy access** and reduce delays in managing a loved one's care. It would also strengthen privacy protections and data integrity by ensuring that only authorized individuals can access sensitive health information. HealthLX is well-positioned to support this vision by enabling identity-verified, FHIR-based access through trusted credentialing partners.

c. What are the potential downsides?

While digital identity credentials offer strong security and access benefits, there are potential downsides. The **enrollment process can be burdensome**, especially for older adults, individuals with limited digital literacy, or those without reliable internet or acceptable forms of ID.

Over-reliance on third-party credentialing services may also raise **privacy concerns**, particularly if patients are uncomfortable sharing personal information or biometric data. Inconsistent implementation across platforms could lead to **fragmented user experiences**, while lack of inclusive design may unintentionally **exclude underserved populations**. To mitigate these risks, CMS should prioritize equitable access, transparency, and interoperability in digital identity policy.

d. How would encouraging the use of CSPs improve access to health information?

Encouraging the use of Credential Service Providers (CSPs) would improve access to health information by enabling **secure, seamless, and scalable digital identity verification** across systems. A trusted, reusable credential allows patients and caregivers to access health records, authorize apps, and manage care across

providers—without repeated logins, paperwork, or in-person verification. This reduces barriers, especially for those navigating complex or multi-provider care journeys. Standardized identity credentials also support **proxy access and consent management**, improving caregiver involvement.

By streamlining identity workflows, CSPs can help eliminate technical friction while ensuring privacy and compliance.

e. What role should CMS/payers, providers, and app developers have in driving adoption?

Driving adoption of digital identity credentials requires coordinated effort across CMS, payers, providers, and app developers:

- **CMS** should lead by **endorsing trusted credentialing frameworks**, integrating digital identity into federal programs (e.g., MyMedicare.gov), and funding outreach to educate patients—especially those in underserved communities.

- **Payers and providers** should **embed credentialing prompts into patient portals, EOBs, and onboarding workflows**, making identity verification a standard part of care navigation.

- **App developers** should support **seamless integration with certified CSPs**, ensuring patients can easily authenticate and access data within third-party tools without friction.

Together, these stakeholders can normalize secure digital identity use, reduce barriers to access, and build public trust in interoperable, patient-centered health data ecosystems.

f. How can CMS encourage patients to get digital identity credentials?

CMS can encourage patients to obtain digital identity credentials by embedding **clear prompts and incentives** across high-traffic touchpoints such as **MyMedicare.gov, EOBs, enrollment materials**, and **telehealth platforms**. Educational campaigns should use **plain language and trusted messengers**—such as providers, pharmacies, and community organizations—to explain the benefits of digital credentials for accessing health records and managing care.

CMS can also **pre-integrate credentialing with account creation** and offer **support for low-digital-literacy users**, including step-by-step guides, live assistance, and mobile-friendly options. By reducing complexity and increasing visibility, CMS can normalize secure digital identity as a key to personalized, accessible care.

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate. (printed page 21038)

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

A: To encourage provider adoption of approved digital health products—especially in rural or resource-constrained settings—CMS and its partners should focus on **reducing implementation burden, building trust in approved tools, and aligning incentives** with clinical workflows and value-based care goals.

Recommended Strategies:

1. **Simplify Integration with EHRs and Workflows**
   - Support **plug-and-play FHIR integration** that minimizes disruption and allows digital tools to operate within the provider's existing systems.
   - HealthLX enables this by offering standards-based APIs and configurable workflows that reduce the lift required by provider IT teams.
2. **Provide Financial Incentives and Technical Assistance**
   - Offer grants, shared savings bonuses, or ACO alignment incentives for digital tool adoption, especially for **rural or safety-net providers**.
   - CMS could pair this with **implementation support programs**, similar to previous meaningful use assistance efforts.
3. **Create a Trusted, Pre-Vetted Marketplace of Tools**
   - Expand access to a **CMS-endorsed registry** of approved, standards-compliant digital health products that meet security, usability,

and outcome impact criteria—giving providers confidence in their value and safety.

4. **Promote Use in Quality and Reporting Programs**
   - Encourage or require integration of approved digital tools in **MIPS, ACO REACH, or other CMS quality programs**, making digital health adoption part of measurable provider performance.
5. **Engage Local Champions and Community-Based Networks**
   - Work with **local provider associations, FQHCs, and HIEs** to identify champions who can promote successful implementations and reduce resistance.

a. What are the current obstacles?

Providers, especially those in rural or community-based settings, face several persistent obstacles to adopting approved digital health tools for their patients:

1. Limited Technical Infrastructure and IT Support
Many smaller practices and rural providers lack the **technical resources**—such as broadband access, modern EHR capabilities, or in-house IT teams—needed to implement and maintain digital health tools.

2. Workflow Disruption and Complexity

Digital health tools often require **manual workarounds or duplicate entry** due to poor EHR integration, disrupting clinical workflows and adding to administrative burden.

3. Unclear ROI or Incentive Alignment

Providers may not see a direct **financial or clinical return** from adopting digital tools, especially when reimbursement models don't support their use or when they're outside of existing quality program incentives.

4. Lack of Trust or Awareness

There's limited awareness of which digital tools are **validated, secure, and standards-compliant**. Providers may be cautious due to privacy, liability, or data-sharing concerns.

5. Burden of Regulatory Compliance

Even approved tools can introduce new **privacy, security, and reporting obligations**, making providers wary of added compliance risks.

b. What information should providers share with patients when using digital products in the provision of their care?

When using digital health products in patient care, providers should clearly explain the purpose of the tool, how it supports clinical decision-making or care coordination, and what types of data will be collected or shared. Patients should be informed about privacy protections, including how their information is secured, who will have access, and how it complies with HIPAA and other regulations.

Providers should also communicate patients' rights to give or withdraw consent, and provide clear instructions on how to access, use, and seek support for the tool.

This transparency builds trust, supports informed participation, and helps ensure responsible use of digital health technologies.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

When recommending a digital health product, providers have a responsibility to ensure the tool is **clinically appropriate, secure, and accessible** for the patient. This includes verifying that the product is **approved or vetted** (e.g., standards-compliant, privacy-protected), clearly explaining its intended use, and integrating it meaningfully into the patient's care plan.

Providers should also confirm that the patient **understands how to use the product**, consents to its use, and knows how their data will be shared or used. Additionally, providers should monitor the product's role in care delivery—reviewing outputs where relevant and adjusting care as needed. HealthLX supports these responsibilities by ensuring digital tools integrated through our platform meet interoperability, consent, and data security standards that align with clinical and regulatory best practices.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Several persistent obstacles limit the development and effective use of digital applications that support physician workflows like quality measurement, documentation, and billing. Chief among them are lack of interoperability, with many EHRs and vendor systems operating in silos, and inconsistent data standards, which hinder accurate quality reporting and clinical automation. Additionally, clinicians face workflow fragmentation, where digital tools operate outside of their primary systems, increasing administrative burden. Financial disincentives, vendor lock-in, and insufficient support for integration also restrict adoption.

To mitigate these issues, CMS can accelerate the adoption of FHIR-based APIs, expand alignment with USCDI and value-based care programs, and incentivize vendors to offer open, standards-compliant access to key data elements. HealthLX helps address these barriers by offering a FHIR-native integration platform that simplifies data access, automates workflows, and enables seamless reporting and billing across systems—making innovation more deployable, sustainable, and scalable in real-world clinical environments.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

It is critically important—particularly for **care equity and quality in both urban and rural settings**—that all data stored in an EHR system be accessible for exchange, regardless of its format (structured or unstructured). Incomplete or inaccessible data introduces care delays, administrative burden, and health disparities, especially for patients receiving care from multiple facilities or in low-resource regions.

a. Current challenges in accessing different data formats.

Providers often face difficulty accessing non-standard data types such as **scanned PDFs, faxed documents, free text notes, or siloed lab results**. These files may be stored in separate modules, lack metadata for indexing, or be excluded from FHIR/API-based data exchange—making them hard to retrieve, integrate, or interpret in time-sensitive workflows.

b. Impact on patient care quality.

Inaccessible data leads to **incomplete clinical context**, which increases the risk of medication errors, redundant testing, delayed diagnoses, and misinformed treatment decisions. This burden is **amplified in rural areas**, where care continuity depends heavily on shared records across unaffiliated providers.

c. Technical barriers to full data accessibility.

Many legacy EHR systems were not built with **interoperability or data normalization in mind**, making it difficult to extract or convert older file types. Additionally, **natural language processing (NLP)** and optical character recognition (OCR) technologies—needed to parse free text or scanned documents—are still maturing and may lack clinical specificity or accuracy.

d. Cost or privacy implications of making all data formats interoperable.

Making all EHR data interoperable involves **infrastructure upgrades, data normalization tools, and advanced security controls**, which can be costly for rural or small providers. There are also **privacy risks** when processing unstructured data, as sensitive information may be embedded in formats that are harder to de-identify or redact.

e. Priority level compared to other interoperability needs.

This is a **high-priority issue**, particularly where **clinical decision-making depends on complete patient histories**, such as in chronic care, emergency medicine, or behavioral health. However, it should be pursued **in parallel** with broader efforts like FHIR-based real-time exchange, payer-provider integration, and patient access initiatives.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

To enable third-party digital products to fully support administrative workflows—such as auto-populating intake forms, accessing provider directories and schedules, and managing appointments—CMS and its partners should prioritize expanding

standards adoption and updating policies to enforce open access to core operational data. Key recommendations include:

- **Broaden Adoption of FHIR Administrative Resources**
  - Widespread implementation of FHIR resources like Schedule, Slot, Appointment, HealthcareService, and Questionnaire is essential to support structured access to provider availability, service details, and digital intake forms.
- **Update CMS API Mandates to Include Operational Data**
  - Current mandates focus primarily on clinical and claims data. CMS should expand requirements to ensure that provider organizations expose administrative endpoints for scheduling and patient intake, using certified APIs.
- **Support Write-Back Capabilities**
  - Many third-party tools can read EHR data, but lack permissions to submit or modify appointment requests, intake forms, or referrals. Enabling secure, standards-based write access—such as through OAuth 2.0 scopes and SMART on FHIR protocols—would improve workflow automation and patient engagement.
- **Align Privacy and Consent Policies for Administrative Data**
  - Clarify HIPAA and CMS guidelines to ensure third-party apps can access non-clinical data with appropriate patient authorization, reducing hesitancy from providers or developers concerned about compliance risks.
- **Encourage Vendor Interoperability Through Certification and Incentives**
  - CMS can accelerate access by requiring certified health IT vendors to implement administrative FHIR resources and integrate appointment workflows into API ecosystems used by approved digital tools.

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

a. Patient Access API.

- **Transaction Model:** FHIR R4-compliant API using OAuth 2.0 authorization.
- **Use Case:** Allows patients to access their clinical and claims data through third-party applications.
- **Query Type:** Individual queries.
- **Constraints:** Access limited to authenticated users with verified identity credentials; provider system participation varies by EHR.

b. Standardized API for Patient and Population Services.

- **Transaction Model:** FHIR-based API leveraging Patient, Encounter, Condition, Observation, and Coverage resources.
- **Use Case:** Supports both patient-specific and population-level data retrieval for care coordination, analytics, and value-based care.
- **Query Type:** Supports both individual and bulk queries.
- **Constraints:** Varies depending on payer or provider participation in bulk export workflows and backend system support for $export.

c. Provider Directory API.

- **Transaction Model:** FHIR-based API using Practitioner, Organization, Location, and HealthcareService resources.
- **Use Case:** Exposes network directory information to support provider search, network status, and care navigation.
- **Query Type:** Individual queries for member-facing search and bulk queries for directory syncing.
- **Constraints:** Data quality and completeness depend on upstream payer or provider data feeds.

d. Provider Access API.

- **Transaction Model:** Secure FHIR API with OAuth 2.0 and role-based access controls for clinical system users.
- **Use Case:** Enables provider-facing applications to retrieve patient data for care coordination and decision-making.
- **Query Type:** Primarily individual patient queries.
- **Constraints:** Role-based access must be clearly defined and managed per provider organization.

e. Payer-to-Payer API.

- **Transaction Model:** FHIR-based data exchange using the Patient, Coverage, and ExplanationOfBenefit resources.
- **Use Case:** Transfers patient health history between payers during transitions or new enrollments.
- **Query Type:** Supports both individual and batch processes.
- **Constraints:** Identity matching, consent, and enrollment verification must be managed prior to data release.

f. Prior Authorization API.

- **Transaction Model:** Based on the Da Vinci Coverage Requirements Discovery (CRD) and Prior Authorization Support (PAS) implementation guides.
- **Use Case:** Automates the prior authorization process through real-time eligibility, documentation support, and authorization submission.
- **Query Type:** Individual service-level transactions.

- **Constraints:** Dependent on payer system readiness and prior authorization logic exposure via API.

g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries?

- **Transaction Model:** FHIR $export operation with Group ID filtering for defined populations.
- **Use Case:** Enables data extraction for risk stratification, population health, and value-based care initiatives.
- **Query Type:** Bulk transactions.
- **Constraints:** Requires backend support for Group resource creation and reliable patient attribution lists.

h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?

- **Support:** Yes, both EHR-launched and standalone app workflows are supported.
- **Application Deployment Process:** Apps are registered through a developer portal with metadata, redirect URIs, scopes, and token configurations. OAuth 2.0 authorization with PKCE is enforced.
- **Constraints:** App approval and integration vary depending on the host EHR platform's security and governance policies.

i. CDS Hooks (for clinical decision support integrations).

- **Support:** Supported in alignment with Da Vinci Clinical Decision Support use cases.
- **Use Case:** Integrates decision support into clinical workflows using contextual triggers (e.g., medication ordering, diagnosis entry).
- **Transaction Model:** RESTful call-and-response model using pre-defined CDS hooks.

- **Constraints:** Effective use depends on EHR system support for hook registration and workflow integration points.

HealthLX's Role:
As an interoperability platform technology vendor, HealthLX enables these capabilities by offering a **FHIR-native and Legacy system interoperability platform** to assist organizations in the transition to a future state FHIR-First future. We see the **broader challenge of the needed configurability of all platforms** that integrate with both payer and provider systems. We believe that the handling of **data normalization, consent management, identity resolution, and transaction orchestration** across these workflows is required to ensure secure and standards-compliant interoperability.

PR-6. Is TEFCA currently helping to advance provider access to health information?

TEFCA is beginning to lay the foundation for **secure, nationwide, interoperable data exchange**, but its **real-world impact on provider access** is still emerging. As Qualified Health Information Networks (QHINs) onboard and technical infrastructure matures, TEFCA is expected to significantly streamline provider access across systems and care settings.

a. Please provide specific examples.

While broad adoption is still in progress, early QHINs like **CommonWell Health Alliance** and **eHealth Exchange** have begun implementing TEFCA-aligned exchanges. These early use cases demonstrate **improved provider-to-provider data access** across EHR vendors and regions, reducing duplication of tests and improving care coordination—particularly during care transitions or emergency visits where cross-network access is vital.

b. What changes would you suggest?

To accelerate provider benefit, we recommend:

- **Expanding support for FHIR-based transactions**, not just document exchange (C-CDA).

- **Accelerating Individual Access Services (IAS)** to ensure providers can assist patients in retrieving their records across QHINs.

- **Clarifying integration pathways for provider organizations** that rely on intermediaries, HIEs, or health IT vendors to participate in TEFCA.

- Providing **financial and technical support** for smaller or rural providers to connect and exchange via TEFCA.

c. What other options are available outside of TEFCA?

While TEFCA is promising, several existing channels currently support provider access:

- Vendor networks such as Carequality and CommonWell, which already enable cross-EHR clinical data exchange.

- HIEs (Health Information Exchanges) that support regional provider access to lab results, imaging, and care summaries.

- FHIR APIs mandated under CMS and ONC rules, which allow authorized providers to access clinical and administrative data at the point of care.

- Direct Trust messaging for secure provider-to-provider communication, though it lacks structured interoperability.

d. Are there redundant standards, protocols or channels or both that could be consolidated?

There are multiple overlapping data exchange standards and protocols that increase complexity for providers:

- **C-CDA vs. FHIR**: Standardizing on **FHIR** for clinical and administrative workflows would reduce duplication and improve data usability.
- **Multiple directory and credentialing systems**: Consolidating provider directories and identity management systems would simplify secure access.
- **Parallel regional and national networks**: Aligning **HIEs with TEFCA** rather than maintaining siloed infrastructures could improve efficiency and reduce cost.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

To ensure providers are positioned to share high-quality, timely, and comprehensive healthcare data—without added burden—CMS should focus on incentive alignment, workflow integration, standards enforcement, and infrastructure support.

1. Align Interoperability with Existing Incentives and Quality Programs

Providers are already motivated to optimize for value-based care, quality measurement (e.g., MIPS, ACO REACH), and care coordination.
CMS can embed FHIR-based interoperability requirements into these programs to:

- Reward timely data sharing through existing quality metrics or bonus structures.
- Enable re-use of structured data already being collected for HEDIS, eCQMs, or risk adjustment.
- Use existing attestation or performance improvement programs to track interoperability participation.

2. Reduce Technical Burden Through Middleware and Certification

Many providers—particularly in rural or small practices—lack the resources to manage direct API or standards implementation. CMS should:

- Support intermediary platforms like HealthLX to handle FHIR normalization, consent, identity matching, and secure exchange.
- Require that certified health IT systems offer "plug-and-play" API integration for digital tools, reducing EHR customization demands.

3. Fund Scalable Infrastructure and Technical Assistance

CMS could provide targeted funding or grants for:

- Upgrading legacy systems to support FHIR R4, USCDI v3+, and TEFCA participation.

- Implementing bulk data access for population-level analysis tied to value-based contracts.
- Offering technical support hubs modeled after past Meaningful Use regional extension centers.
- Financially support the open-source nature of the technical specifications that should be shared with all stakeholders within the ecosystem that have mandates to meet CMS regulations

4. Support Workflow-Friendly Data Exchange Models

Interoperability initiatives must integrate with, not disrupt, clinical workflows. CMS can:

- Promote tools that auto-populate documentation, intake forms, or referrals using shared data.
- Use SMART on FHIR and CDS Hooks to surface decision support and quality reminders at the point of care.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

To reduce the administrative burden of clinical quality reporting while improving data utility, CMS and its partners should prioritize **standards-based automation, data reuse, and real-time interoperability infrastructure**.

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports? **Reduces provider burden** by eliminating manual chart abstraction and enabling centralized, standardized data extraction.

Benefits:

- Supports **automated, scalable quality measurement** at the CMS level using EHR-native data.
- Promotes **data reuse** for multiple CMS programs (e.g., ACO REACH, MIPS, MA Stars) from a single export process.

- Enables **population-level analytics** for both retrospective and prospective quality improvement.

Downsides:

- Requires EHR vendors to fully support **Bulk FHIR $export operations** and the **Group resource** for patient attribution—features not yet universally implemented.
- Potential **data quality issues** if clinical documentation is inconsistent, unstructured, or incomplete.
- Security and privacy concerns around **large-scale data movement** must be carefully managed.

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

- CMS can consolidate efforts by aligning **quality reporting requirements with existing interoperability initiatives**, such as FHIR-based APIs and TEFCA participation.
- Leverage clinical data already required for **public health reporting, care coordination, and payer APIs** to populate eCQMs and other metrics.
- Enable **dual-purpose use of shared data** through tools like SMART on FHIR apps and CDS Hooks, which can both support clinical care and quality metric collection at the point of care.
- Align data requirements across programs (e.g., MIPS, HEDIS, eCQMs) using **USCDI and FHIR resources** to avoid duplicative data entry.

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

To support more efficient digital quality measurement and inform care in real time, CMS should consider requiring that data registries:

- **Support FHIR APIs and Bulk FHIR exports**, including structured access to all resources used in eCQMs.
- Enable **bi-directional data flow**, so registries can return quality measure feedback and gaps-in-care alerts to providers.
- Provide **near real-time dashboards** for providers to monitor progress on metrics, not just retrospective summaries.
- Standardize **attribution logic, patient identity matching, and data provenance tracking** to ensure accurate measurement and trust in reported data.
- Integrate with **clinical decision support systems**, using registry data to improve in-the-moment care quality.

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

CMS can play a leading role in modernizing identity workflows by encouraging providers to accept trusted digital identity credentials—such as those from Login.gov, ID.me, or CLEAR—instead of requiring separate logins per provider organization. A unified, patient-managed credential enhances both security and usability, supporting seamless access to care and digital health tools.

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

To transition away from proprietary logins, providers would need:

- Technical integration support to link EHR portals, patient-facing tools, and identity workflows with certified Credential Service Providers (CSPs).
- Clear CMS guidance and policy alignment encouraging or requiring the acceptance of NIST 800-63-3 IAL2/AAL2-compliant credentials.
- Intermediary platforms like HealthLX to help translate digital identity assertions into provider-trusted authentication events—reducing local IT burden.
- Training and trust-building materials to help providers and staff understand the security and compliance benefits of shared digital credentials.

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

CMS can balance privacy and access by:

- Requiring that **CSPs follow strict privacy, consent, and data minimization practices**, ensuring identity is verified without over-collection of sensitive information.
- Promoting **patient-controlled data access**, where identity credentials are used to authorize specific, limited access scopes—e.g., for digital tools, caregivers, or payers.
- Offering **opt-in options and transparency** around what information is shared, with whom, and for how long—reinforcing trust and informed consent.
- Supporting **proxy and caregiver workflows** through delegated identity models that respect privacy while enabling critical care coordination.

Digital identity should serve as a **gateway to trusted access**, not a barrier. By supporting interoperability and patient empowerment, CMS can improve access while maintaining privacy, equity, and compliance.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers?

- Providers may face **integration complexity** when connecting existing systems with external Credential Service Providers (CSPs).
- There may be **workflow disruption** or reluctance from staff unfamiliar with digital identity models.
- **Managing access delegation** (e.g., caregivers, administrative staff) through verified credentials adds complexity in high-volume environments.

Benefits:

- Reduces **password management overhead**, fraud risk, and identity-related errors.

- Enhances **security, auditability, and compliance**, especially when dealing with sensitive data access.
- Supports **faster onboarding** of patients and external partners (e.g., care coordinators, digital tools) using verified identities.
- Creates a foundation for **seamless, scalable interoperability** across systems and regions.

b. How would requiring their use improve access to health information?

Mandating digital identity credentials would standardize how patients, providers, and third-party apps authenticate, enabling consistent, secure access to health information regardless of location or EHR. It simplifies cross-provider data sharing, enhances patient-directed exchange, and reduces the reliance on disjointed, portal-specific credentials—especially helpful for caregivers managing multiple patient relationships.

For providers, it streamlines onboarding for external apps, prior auth tools, and registry reporting, promoting a unified digital health ecosystem.

c. What are the potential downsides?

- May be burdensome for smaller practices or those with limited IT resources.
- Risk of excluding patients or providers without easy access to credentialing tools (e.g., rural areas, limited ID access, language barriers).
- Could introduce workflow changes that require retraining, particularly around proxy access and digital consent.
- Relies on trust in third-party CSPs, which may require ongoing oversight and accreditation.

d. What impact would mandatory credentials have on a nationwide provider directory?

Mandatory credentials could greatly enhance the accuracy, security, and usability of a national provider directory by:

- Ensuring all provider entries are tied to a verified, authenticated identity, reducing duplicates and impersonation.

- Enabling role-based access to update directory entries (e.g., credentialed providers updating NPI-linked data).
- Supporting federated identity models that allow provider access to shared infrastructure without re-authentication.
- HealthLX could integrate identity credentials into provider directory APIs to support real-time updates, access tracking, and trust signals for downstream systems.

e. How could digital identity implementation improve provider data flow?

With trusted identity credentials, providers could:

- Streamline access to external data systems (e.g., registries, payer APIs, TEFCA QHINs) without redundant credentials.
- Enable secure, standards-based access control, improving data governance and reducing administrative overhead.
- Facilitate automated audit trails and accountability, especially for sensitive data flows like behavioral health or SDOH information.
- HealthLX supports credential-based access enforcement, ensuring that only authorized providers or apps retrieve data via FHIR APIs or consent-managed workflows.

f. Would combining FHIR addresses and identity improve data flow?

Yes. Combining verified digital identity with FHIR endpoint addresses would:

- Enable precise, scalable routing of data to verified endpoints.
- Eliminate ambiguity in cross-system communication, supporting automated workflows (e.g., prior auth, care transitions, referrals).
- Improve patient safety by reducing misdirected or fragmented data delivery.

HealthLX Recommendation:
CMS should support a national registry of FHIR endpoints tied to verified digital identities (e.g., NPI + credential), enabling smart routing and reducing integration complexity across the digital health ecosystem.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

Members of trust communities such as QHINs, participants, and subparticipants under TEFCA can play a central role in reducing provider and patient burden while enhancing identity management and security by standardizing credentialing processes, streamlining access, and enabling interoperability across systems.

Key Opportunities:

1. Adopt and Promote Federated Identity Models
2. Trust communities should support reuse of IAL2-compliant digital credentials (e.g., Login.gov, ID.me) across networks—enabling patients and providers to authenticate once and access multiple systems securely, without redundant logins or re-verification.
3. Centralize and Share Trusted Identity Assertions
4. By leveraging the TEFCA ecosystem, QHINs can act as trusted identity brokers, sharing verified digital identity assertions across participants in a standardized, privacy-preserving manner. This reduces friction in cross-network data exchange.
5. Enable Seamless Role-Based Access Control (RBAC)
6. Support for structured identity data allows trust community members to implement role-aware access policies, streamlining provider onboarding and enabling delegated access for caregivers, proxies, or clinical staff.
7. Integrate Identity with Workflow and Endpoint Directories
8. Linking verified identities to FHIR endpoints, care team roles, and provider directories can dramatically reduce administrative effort in configuring and securing data exchange.
9. Coordinate on Consent and Privacy Policy Enforcement
10. Standardizing consent management across trust communities based on verified identity ensures secure, patient-directed access, reducing legal ambiguity and compliance overhead.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

Yes, ASTP/ONC should consider revising certain information blocking exceptions under 45 CFR part 171 to better align with the goals of broad EHI access, patient empowerment, and market competition. While the exceptions were designed to protect against undue risk, overly broad or ambiguous language can unintentionally enable data hoarding, delay innovation, and sustain vendor lock-in.

Recommended Areas for Revision:

- **Reevaluate the "Infeasibility" Exception**
  - Current Issue: This exception is often cited to deny access based on vague claims of technical limitations or resource constraints.
  - Recommendation: Narrow the scope by requiring documented, time-bound remediation plans and technical evidence if invoked. Clear criteria would prevent misuse while still protecting against genuine limitations.

- **Refine the "Security" Exception**
  - Current Issue: Some actors invoke security as a blanket reason to block access, even when the requesting party meets equivalent security standards.
  - Recommendation: Require that denial based on security risks be objectively justified and proportionate, and aligned with NIST-aligned CSPs and FHIR API frameworks already in use across regulated environments.

- **Clarify "Content and Manner" Exception Boundaries**
  - Current Issue: This exception can be used to avoid API-based access by offering only outdated formats (e.g., PDFs, CCDAs).
  - Recommendation: Prioritize standards-based (e.g., FHIR) exchange as the default, and define clear expectations for timely adoption of modern formats under this exception.

- **Strengthen Enforcement Transparency and Timelines**
    - Include public reporting of enforcement trends and expected response timelines to reduce delays in resolution and promote voluntary compliance through visibility.

Updating the exceptions will clarify intent, promote good-faith participation, and accelerate equitable data access across the healthcare ecosystem.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

For provider categories not currently subject to CMS-established information blocking disincentives under 42 U.S.C. 300jj(3), the most effective approach would be to align disincentives with existing CMS programs that directly influence provider behavior—without imposing undue burden or harming access to care. The goal should be to incentivize compliance through accountability while preserving support for providers that may face legitimate resource constraints.

Recommended Disincentive Models:

Participation-Linked Disincentives (Soft Enforcement)

- Tie compliance with information blocking rules to eligibility or preferential scoring within programs like ACO REACH, MIPS, Medicaid Promoting Interoperability, or future value-based initiatives.
- Example: Noncompliant providers may receive reduced quality scores or reduced eligibility for shared savings bonuses.

Public Transparency ("Naming and Shaming")

- Publish lists of organizations with verified violations of information blocking rules, particularly when no security or technical justification is documented.
- This enhances market pressure and reputational accountability without requiring direct financial penalties.

Data Access Restrictions

- Limit access to federal data-sharing infrastructure, such as TEFCA QHIN participation or CMS APIs, for repeat violators. This aligns access privileges with good-faith participation in interoperability efforts.

Progressive Corrective Action Plans

- For smaller or resource-constrained providers, CMS could implement graduated corrective actions before applying financial penalties. These may include technical assistance, documentation audits, or mandated remediation plans.

Providers should be held accountable for willful obstruction of data access, but enforcement must also distinguish between deliberate information blocking and systemic or resource-related limitations. CMS should prioritize standards-based enablement (e.g., FHIR APIs, identity integration) alongside smart enforcement mechanisms that drive meaningful progress.

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

CMS can encourage providers to report information blocking to the ONC's Information Blocking Portal by **embedding awareness and easy access** into existing provider touchpoints—such as **Medicare program portals, MIPS dashboards, EHR interfaces, and onboarding materials** for interoperability programs. Many providers remain unaware that they can report blocking behavior by EHR vendors, health systems, or networks.

CMS could also work with provider associations and health IT implementers to **promote education, simplify reporting processes**, and clarify what constitutes a valid complaint.

Encouraging provider reporting would likely have a **positive impact on data exchange** over time. It would:

- **Highlight systemic barriers** (e.g., lack of API access, data export restrictions) that obstruct patient care or provider collaboration.

- **Hold vendors and networks accountable**, especially in cases where market control limits interoperability progress.

- **Strengthen enforcement efforts** under ONC by providing real-world insights into common compliance failures.

However, to avoid unintended negative impacts—such as **misuse, fear of retaliation, or reporting fatigue**—CMS should pair the process with **confidentiality protections, guidance on complaint quality**, and **feedback loops** to show how reporting leads to meaningful outcomes.

D. Payers

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

TEFCA represents a major step toward national interoperability, but several **policy and technical limitations** could hinder payer participation—particularly in terms of scalability, data richness, and alignment with payer-driven workflows. To ensure broad adoption and impact, CMS and ONC should address the following key areas:

1. Technical Limitations

- **Document-Centric Exchange Model**

    - TEFCA currently prioritizes **C-CDA-based document exchange**, which limits **data granularity, automation, and analytics**. This model is not well-suited for payers that need structured, computable data for quality measurement, utilization review, and member engagement.
    - **Suggested Change:** Translate to FHIR resources C-CDA structured documents in order to integrate **FHIR-based query support** into TEFCA's exchange protocols—particularly for clinical and administrative workflows used by payers.

- **Lack of Bulk Data Support**

- TEFCA does not yet support **FHIR Bulk Data ($export)** operations needed for population health and risk management.
- **Suggested Change:** Prioritize roadmap alignment with **Da Vinci and CARIN implementation guides** to support **population-level data retrieval** through QHINs.

- **Limited Support for Write-Back Workflows**

  - TEFCA is largely **read-only**, which limits its utility for bidirectional workflows (e.g., prior authorization, care plan updates, or referrals).
  - **Suggested Change:** Expand TEFCA to support **write transactions** under well-defined security and consent models. We believe this could be accomplished but only after adoption of TEFCA has been achieved.

2. Policy Limitations

- **Ambiguity Around Payer Roles**

  - The definition and scope of a **payer as a QHIN participant or subparticipant** is still evolving. Many payers are unsure how TEFCA participation will overlap or conflict with **existing API mandates** or private networks (e.g., Carequality, CommonWell).

  - **Suggested Change:** Clarify policy alignment between **TEFCA and CMS interoperability rules (e.g., Patient Access API, Payer-to-Payer API)** to avoid duplication or contradiction.

- **Cost and Onboarding Burden**

  - Many payers—especially Medicaid MCOs or regional plans—lack the technical and financial resources to participate directly in TEFCA without **intermediary or shared infrastructure models**.
  - **Suggested Change:** CMS should consider **financial incentives, grants, or technical assistance programs** to support payer onboarding and participation, especially in underserved markets, but only do so with an approach to support universal technical data mapping requirements to assure both semantic clarity across payers and a means to test for conformance. Otherwise, costs will be mismanaged on a stakeholder by

stakeholder basis.

3. Potential Hindrance to Participation

If these issues are not addressed, payer participation in TEFCA may be:

- **Limited to large national plans** with the infrastructure to support parallel exchange models.
- Delayed by **confusion over value proposition**, especially when private HIEs or API-based approaches offer faster, more tailored access.
- **Undermined by lack of functionality** for core payer needs like quality reporting, prior authorization, and value-based contracting.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, (printed page 21039) providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

To accelerate the implementation and meaningful use of APIs across the healthcare ecosystem, CMS should take a multi-pronged approach that combines incentives, enforcement, technical guidance, and ecosystem alignment—building on the momentum of Blue Button 2.0, Patient Access API, and Data at the Point of Care (DPC) efforts.

1. Tie API Adoption to Value-Based Payment Programs

CMS should integrate API usage as a performance metric or participation requirement within value-based programs such as ACO REACH, Medicare Advantage Stars, and Medicaid managed care contracts. This reinforces interoperability as a strategic lever, not just a compliance obligation.

2. Expand Financial Incentives and Technical Assistance

Smaller or resource-constrained payers may lack the infrastructure to meet FHIR and security standards without support. CMS can:

- Provide grants or shared infrastructure support for Medicaid MCOs and regional plans.

- Fund FHIR accelerators and implementation playbooks tailored to payer-provider data exchange and cross-payer coordination.

3. Promote Standardized, Pre-Vetted Implementation Guides

Adoption is slowed by variability and uncertainty. CMS can help by:

- Requiring or recommending adherence to specific HL7® implementation guides, such as those from CARIN (Patient Access), Da Vinci (Prior Auth, Payer-to-Payer), and Gravity Project (SDOH).
- Supporting the technical specifications underpinning all stakeholders to meet CMS requirements.
- Supporting conformance testing and certification programs for payers and vendors to ensure real-world API readiness.

4. Publicly Showcase API Utilization and Compliance

Following the Blue Button 2.0 model, CMS can improve transparency and accountability by:

- Publishing payer-specific API implementation statuses, usage metrics, and compliance scores.
- Recognizing leading payers through public dashboards, API quality awards, or CMS partnership features.

5. Streamline Integration Across Use Cases

- CMS should support alignment of APIs across key workflows—such as prior authorization, quality reporting, and patient data access—by requiring reusable infrastructure (e.g., SMART on FHIR app launch, OAuth 2.0 authentication) and multi-role access models that serve patients, providers, and payers through a shared framework.

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

CMS can play a pivotal role in accelerating payer acceptance of trusted digital identity credentials (e.g., ID.me, CLEAR, Login.gov) by embedding them into policy requirements, technical standards, and member engagement workflows. Replacing proprietary logins with NIST 800-63-3 IAL2/AAL2-compliant credentials enhances both security and interoperability, while reducing patient and caregiver burden.

Key Strategies:

- **Include Digital Identity Standards in Interoperability Regulations**
    - Require or incentivize payers—especially those participating in Medicare Advantage, Medicaid managed care, or ACA marketplaces—to accept at least one federated credential from a CMS-approved list of Credential Service Providers (CSPs).
    - Update Patient Access API and Provider Access API specifications to support identity federation through OAuth 2.0 + OpenID Connect, aligned with SMART on FHIR profiles.

- **Incentivize Identity Integration Through Program Participation**
    - Tie adoption of federated digital identity workflows to Stars Ratings, value-based care scoring, or administrative simplification incentives, especially where identity management intersects with consent, care navigation, and claims authorization.

- **Provide Technical Implementation Guidance and Sandbox Support**
    - Offer CMS-sponsored toolkits, sample code, and sandbox environments to help payers integrate with CSPs securely and efficiently.
    - Fund pilot programs to demonstrate real-world implementations across Medicaid, Medicare Advantage, and dual-eligible populations.

- **Align with TEFCA and QHIN Identity Assurance Requirements**
    - Reinforce that digital identity credentials accepted by payers will facilitate participation in TEFCA networks, where IAL2-level identity is required for patient access services.

- Encourage interoperability between TEFCA-compliant trust frameworks and payer-facing identity verification processes.

- **Support Member Education and Caregiver Access Use Cases**
  - Equip payers with CMS-approved communication materials to explain the benefits of digital identity to members and caregivers—especially for secure, proxy-enabled access to health data.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

A nationwide provider directory that includes **FHIR endpoints** and is linked to **verified digital identity credentials** would offer substantial value to payers by improving **network accuracy, care coordination, data exchange, and administrative efficiency**. Today's fragmented directories create a significant burden for payers in maintaining current, validated provider information—especially across multi-payer and multi-provider ecosystems.

Key Benefits to Payers:

1. **Streamlined Data Exchange and Routing**

   - Associating FHIR endpoints with **digitally verified provider identities** enables **automated, secure routing of clinical and administrative data**, reducing delays and errors in prior authorization, care transitions, and quality reporting workflows.

2. **Improved Provider Network Accuracy**

   - A centralized, trusted directory with verified identity credentials ensures that **provider affiliations, locations, and contact details** are accurate and up to date—supporting **directory transparency compliance** and improving **member search experiences**.

3. **Reduced Administrative and Compliance Burden**

- Consolidating provider identity and contact information into a **federated, standards-based directory** decreases duplicative maintenance, improves reporting for **CMS directory audits**, and reduces the risk of sending sensitive data to invalid or outdated endpoints.

4. **Supports Interoperability and TEFCA Participation**

- A nationwide directory that aligns with TEFCA and FHIR standards would enhance **payer readiness for national exchange**, particularly as TEFCA QHINs begin routing data based on verified identities and endpoints.

5. **Enables Role-Based Access and Delegated Permissions**

- Tying digital identity credentials to FHIR endpoints allows payers to **implement fine-grained access control**, ensuring that only credentialed providers or proxies can initiate or respond to data transactions.

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

Payers have a strategic opportunity to reduce provider burden and enhance data quality by **supporting standards-based, scalable approaches to quality measurement**—particularly through shared infrastructure, aligned metrics, and real-time data exchange.

a. How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?

There is growing interest from both payers and providers in EHR capabilities that support bulk FHIR data extraction, allowing payers to handle quality measurement on behalf of providers.

This model:

- Reduces the reporting burden on provider organizations, especially small or resource-limited practices.
- Improves data consistency and timeliness, as payers can ingest structured data directly from EHRs using FHIR Bulk $export and Group resources.

- Supports retrospective and prospective quality insights, which are harder to achieve when providers rely solely on embedded EHR tools.

b. In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

To reduce redundancy in provider reporting and support real-time clinical quality improvement, payers and CMS can:

- **Align Measure Requirements Across Programs**
  - Harmonize eCQM specifications across CMS programs and private payer contracts using **USCDI, FHIR, and HL7 implementation guides** to enable **data reuse** from a single source of truth.
- **Enable Shared Technology Infrastructure**
  - Payers can invest in **FHIR-native platforms** (like HealthLX) to **ingest, validate, and analyze** quality data for both internal use and submission to CMS on behalf of providers.
- **Support Real-Time Dashboards for Providers**
  - Return quality insights and gaps-in-care alerts to providers through **SMART on FHIR apps**, provider portals, or EHR-embedded tools to improve care delivery, not just reporting compliance.
- **Leverage TEFCA and Trust Frameworks**
  - Encourage the use of TEFCA QHINs or HIEs to deliver **bidirectional exchange of quality-relevant data** and minimize multiple submission endpoints.

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

CMS can encourage payers to submit information blocking complaints to the ONC's Information Blocking Portal by integrating education, accountability, and policy alignment into payer-facing programs and compliance workflows. Many payers experience barriers to data exchange—especially with providers or health IT vendors—but may not realize that these qualify as reportable offenses under 45 CFR Part 171.

- **Strategies for Encouragement:**
    - Embed Reporting Guidance into Regulatory Workflows
    - Include information blocking complaint prompts and education within CMS's oversight of Payer-to-Payer API, Patient Access API, and prior authorization compliance reviews.

- Offer Confidential, Streamlined Submission Tools
    - Simplify the reporting process with payer-specific templates or intake forms, and offer confidentiality protections to encourage submission without fear of vendor retaliation.

- Publicly Recognize Good-Faith Participation
    - Acknowledge payers who proactively report information blocking as part of broader interoperability improvement efforts, particularly in value-based care or Medicaid programs.

- Coordinate with Industry Associations
    - Partner with AHIP, Medicaid Directors, and regional health plans to raise awareness of how and when payers should report blocking behaviors that impede authorized data access.

- Surfacing real-world examples of vendor or provider noncompliance can help ONC target enforcement, clarify ambiguous standards, and ensure accountability in interoperability initiatives.
- Encourages a culture of compliance, transparency, and proactive resolution, improving payer access to critical data for member care coordination, risk adjustment, and quality improvement.

Risks to Mitigate:

- CMS should monitor for overuse or misuse of the reporting mechanism and ensure educational outreach distinguishes between true blocking behavior and legitimate technical or regulatory limitations.

HealthLX's Role:
HealthLX supports payers in overcoming technical data sharing barriers and recognizes when information blocking—intentional or not—undermines access. We promote standards-compliant integration and advocate for policies that foster secure, equitable, and timely data exchange across the healthcare ecosystem.

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

To accelerate innovation for Medicare beneficiaries and caregivers, CMS should focus on **lowering integration barriers, expanding access to data, aligning reimbursement models with digital tools**, and offering **policy clarity and development incentives**. Developers need both technical access and a viable business case to prioritize Medicare-specific solutions.

Short-Term (Next 2 Years):

1. **Expand Real-Time, Standards-Based Data Access**
    - Ensure **FHIR APIs (e.g., Patient Access, Provider Access, and Payer-to-Payer)** are universally adopted and available across all Medicare Advantage and Fee-for-Service plans.
    - Support **open sandbox environments** with synthetic beneficiary data to lower development and testing barriers.

2. **Simplify Digital Credentialing and Authorization**

   - Enable app developers to integrate **digital identity credentials (e.g., Login.gov, ID.me)** into their workflows to support seamless, secure access for patients and caregivers.

3. **Launch a CMS Developer Certification Pathway**

   - Create a **pre-vetted or endorsed digital product registry**, signaling compliance with CMS and ONC standards and increasing developer trust and visibility.

4. **Fund Targeted Innovation Challenges**

   - Sponsor **Medicare-specific digital health accelerators or hackathons** focused on caregiving, chronic disease management, medication adherence, and SDOH navigation.

5. **Clarify CMS Support for App Use in Quality and VBC Programs**

   - Provide **guidance on how third-party tools can be used for MIPS, ACO, and MA Stars reporting**, allowing developers to build directly into these workflows.

6. **Define anticipated three to five year policy roadmap**

   - CMS and ONC regulations have successfully helped to set positive directions of transparency and burden reduction while also driving investment decisions for payers, providers and technology vendors. It would be very beneficial for CMS and ONC to provide a three to five year policy roadmap that highlights anticipated directions and regulations. The roadmap should be updated and published every three to six months to maintain alignment with current priorities. This would allow all stakeholders to align their development roadmaps to support accelerated implementation timelines. It would also help to shift the mindset of payers and providers away from simply addressing a compliance checkbox toward strategic implementation in alignment with regulations.

Longer-Term (3–5+ Years):

1. **Integrate Digital Tools into Core CMS Programs**
    - Allow CMS-certified apps to be **prescribed or recommended by providers and payers**, and **eligible for reimbursement or quality bonuses** when their use improves health outcomes.

2. **Support a Medicare App Ecosystem**
    - Expand Blue Button into a **"Medicare App Store"** framework where trusted, standards-based apps are discoverable by beneficiaries and caregivers.

3. **Facilitate Cross-Network Exchange and Scaling via TEFCA**
    - Ensure TEFCA supports **app-based access, third-party app onboarding**, and integration with QHINs for developer-scaled interoperability.

4. **Expand Coverage for Digital Services**
    - Continue evolving reimbursement models for **remote monitoring, asynchronous communication, and digital coaching** solutions—especially those designed for aging populations.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

To stimulate broader developer engagement and enable richer, patient-centered applications for Medicare beneficiaries and caregivers, CMS should prioritize **expanding the scope and granularity of data available through APIs**, especially beyond the current Blue Button 2.0 (claims-focused) model.

Most Valuable Additional Data Types:

1. **Medicare Advantage (Part C) and Medicaid Encounter Data**

- These data are essential to reflect **complete care histories**, especially for dual-eligible beneficiaries and populations managed under **value-based contracts**.
- Enables developers to build tools for **care navigation, analytics, and outcomes tracking** that are currently limited by claims gaps.

2. **Real-Time Eligibility and Coverage Details**

   - Access to plan-specific **benefits, copays, deductibles, and coverage limitations** supports the development of **cost transparency tools, digital onboarding, and referral guidance apps**.

3. **Prior Authorization Status and Requirements**

   - Enables real-time app notifications and **workflow integration for providers, caregivers, and patients**, improving speed-to-care and reducing administrative friction.

4. **Provider Network Participation and Attribution Data**

   - Identifying in-network providers, PCP assignments, and care team affiliations supports better **navigation, care coordination, and accountability-based applications**.

5. **SDOH, Home Health, and LTC Service Utilization**

   - Data reflecting social needs, community-based service access, and long-term care usage would empower developers to support **caregiver management tools, SDOH intervention apps, and aging-in-place solutions**.

6. **Quality Performance and Risk Scores**

   - Data from Stars ratings, HEDIS, and risk adjustment models can be used to **personalize patient outreach, steer provider referrals**, and build **equity-focused dashboards**.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

The Blue Button 2.0 API provides a strong foundation with Medicare claims, coverage, and demographic data—but to unlock more impactful digital health tools for Medicare beneficiaries and caregivers, developers need access to **complementary data sources that fill clinical, contextual, and operational gaps**.

Top Complementary Data Sources:

1. **Electronic Health Record (EHR) Data from Providers**

    - Includes **clinical notes, lab results, vitals, immunizations, problem lists, and care plans**.
    - Enables a more complete picture of a patient's health status and augments claims-based data with **clinical specificity and timeliness**.

2. **Medicare Advantage (Part C) and Medicaid Encounter Data**

    - Captures services not represented in FFS claims—especially critical for dual-eligible beneficiaries.
    - Supports **risk scoring, quality improvement, and care coordination apps**.

3. **Real-Time Prescription Benefit (RTPB) and Pharmacy Data**

    - Data from **Part D and retail pharmacy sources** enhance medication adherence tools, polypharmacy monitoring, and out-of-pocket cost estimators.

4. **Social Determinants of Health (SDOH) and Community Service Referrals**

    - Includes **housing, transportation, food insecurity, and caregiver support data** from health plans, HIEs, or social service referral platforms.

    - Drives innovation in **aging-in-place solutions, SDOH intervention apps, and caregiver-facing tools**.

5. **TEFCA-Enabled Clinical Networks (via QHINs)**

   - Data from **interoperable health information networks** including hospital discharge summaries, specialist consultations, and longitudinal care records.

   - Enables **care transition and hospital readmission prevention solutions**.

6. **Device, Remote Monitoring, and Patient-Generated Health Data (PGHD)**

   - Data from wearables, home monitoring tools, and mobile apps enhance chronic disease management tools and **caregiver remote oversight**.

7. **Provider Directory and Network Participation Data**

   - Helps identify **in-network, accepting providers**, supporting better navigation tools and real-time referral decision-making.

c. What obstacles prevent accessing these data sources today?

Despite significant progress through CMS's Blue Button 2.0 and interoperability mandates, access to many critical complementary data sources remains limited due to **technical, policy, and operational barriers**. These gaps constrain the development of truly comprehensive and personalized digital tools for Medicare beneficiaries and caregivers.

Key Obstacles:

1. **Technical specifications in the form of a data mapping canonical**
   - This is essential to normalize how data is technically supplied from source systems to a target FHIR Resource definition. Without a universally accepted and endorsed canonical, the achievement of Semantic interoperability will not be achieved.
   - The CMS regulations support the 'what has to be achieved'. The mapping canonical supports the 'How the data needs to be defined' for the what to be achieved. Technical developers have a challenge in

properly mapping the data to the target FHIR Server because the technical specifications live in various Implementation Guides all in a state to flux.
    - All technical specifications will evolve and mature over time. CMS will not have the basis for an audit on semantic alignment in the data / version across the stakeholders unless there is agreement on the meaning of the data being exchanged.

2. **Fragmentation Across Systems and Networks**

    - EHR data, social service data, and device-generated information are spread across **disconnected platforms** with varying access models, data standards, and permissions.
    - TEFCA participation is still limited, and **cross-network exchange** among QHINs is just emerging.

3. **Limited Standardization and FHIR Adoption**

    - Many data sources (e.g., Medicaid, MA, SDOH) still rely on **non-FHIR formats or custom APIs**, which complicate integration and increase development costs for app developers.

4. **Privacy, Consent, and Proxy Access Challenges**

    - Inconsistent consent models, lack of caregiver authorization frameworks, and concerns over HIPAA compliance **slow access to sensitive data**, especially SDOH and behavioral health records.

5. **Lack of Real-Time Data Availability**

    - Blue Button 2.0 and many payer APIs are **claims-based and retrospective**, limiting their utility for real-time clinical decisions or proactive care management.

6. **Inconsistent Payer and Provider Participation**

    - Not all payers—particularly Medicare Advantage or Medicaid plans—fully implement CMS-mandated APIs or expose **complete encounter, prior**

auth, or pharmacy data.

- Provider organizations may restrict EHR access to third-party tools, citing **security concerns or unclear policy interpretation** under information blocking rules.

7. **Cost and Operational Burden**

- Smaller provider organizations and health IT vendors may lack the resources to maintain compliant APIs, consent management systems, or federated identity workflows, especially for multi-source data access.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

To fully unlock the potential of digital health tools for Medicare beneficiaries and caregivers, CMS and ONC should expand their interoperability framework by **formalizing access to additional FHIR-based APIs** that support clinical, administrative, and social care integration. These APIs would address existing access gaps, enable real-time care coordination, and stimulate third-party app innovation.

1. Real-Time Prior Authorization APIs

- Based on the **Da Vinci CRD, DTR, and PAS implementation guides**, these APIs allow providers and digital tools to:
  - Discover coverage requirements for services,
  - Retrieve necessary documentation, and
  - Submit and track prior authorization requests.
- **Why it matters:** Streamlines care approvals, reduces delays, and enables API-based automation across provider, payer, and patient tools.

2. Coverage and Benefits API

- Enables real-time access to **plan-level coverage details**, including benefits, cost-sharing, and service limitations.
- Supports **cost estimators, referral logic, and decision support** in patient- and provider-facing apps.
- Aligns with CMS's Interoperability and Transparency rules but needs broader enforcement and adoption.

3. FHIR Scheduling and Appointment APIs

- Based on HL7 FHIR Schedule, Slot, and Appointment resources.
- Would allow third-party tools to **query availability and book appointments across provider systems**—critical for care navigation apps.
- These APIs should be **required for certified EHRs and provider-facing platforms**.

4. Electronic Referrals and Closed-Loop Care Coordination APIs

- Enable providers and digital tools to initiate, update, and track referrals and care transitions using FHIR Task, ServiceRequest, and CarePlan resources.
- Supports care coordination and reduces **referral leakage**, delays, and follow-up gaps.

5. Social Determinants of Health (SDOH) and Community Resource APIs

- Based on the **Gravity Project's FHIR implementation guides**, these APIs standardize exchange of SDOH assessments, needs, interventions, and referrals.
- Critical for **aging-in-place, caregiver, and social care integration apps**.

6. Quality Reporting APIs

- APIs to retrieve, submit, and receive feedback on quality measure data (e.g., eCQMs, HEDIS).
- Align with efforts to **automate digital quality measurement (dQM)** and enable **real-time feedback loops for providers and apps**.

7. Identity and Consent APIs

- APIs that enable **digital identity validation, delegated access (e.g., caregivers), and consent management**.

- These are foundational for safe, scalable interoperability and would promote **trusted access** across third-party apps, TEFCA networks, and CMS portals.

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

Implementing trusted digital identity frameworks—such as those aligned with **NIST 800-63-3 IAL2/AAL2**—offers transformative potential for secure, scalable, and user-friendly access to health information across payers, providers, patients, and caregivers. However, it also presents several adoption and infrastructure challenges that must be addressed to ensure equity, usability, and interoperability.

Challenges:

1. **Enrollment Complexity and Digital Literacy Gaps**
    - Credentialing processes often require biometric capture, document upload, or multi-step authentication, which can be difficult for **older adults, caregivers, or underserved populations**.
2. **Inconsistent System Integration Across Health IT Vendors**
    - Many provider and payer portals still rely on **proprietary logins or custom identity models**, making it difficult to implement **federated digital identity** consistently across platforms.
3. **Privacy Concerns and Trust Issues**
    - Patients and providers may be reluctant to adopt digital identity solutions involving **government-issued IDs or third-party CSPs**, especially if privacy protections are not clearly communicated.
4. **Proxy and Delegated Access Remain Undersupported**
    - There is no universal model for **caregiver identity delegation**, limiting secure access for spouses, adult children, or legal guardians managing

care.

5. **Cost and Infrastructure Barriers for Smaller Organizations**

   - Implementing identity federation and secure access control mechanisms may be technically and financially burdensome for **rural providers, small payers, and community health organizations**.

Benefits:

1. **Streamlined, Secure Access Across Systems**

   - A single, verified credential enables **patients and providers to access multiple portals and applications** without juggling multiple usernames or re-verifying identity.

2. **Improved Data Exchange and Consent Management**

   - Trusted digital identities support **role-based access, consent logging, and identity verification**—critical for FHIR-based API access, TEFCA participation, and third-party app authorization.

3. **Support for Patient-Directed Exchange and App Ecosystems**

   - Enables scalable, patient-initiated sharing of health data with **digital health apps, caregivers, and community organizations**.

4. **Fraud Prevention and Auditability**

   - Reduces impersonation risks and strengthens **audit trails**, especially important for sensitive use cases like behavioral health, prior authorization, and social services.

5. **Foundation for Nationwide Provider and Patient Access Models**

   - Digital identity enables **TEFCA Individual Access Services (IAS)**, supports public health reporting, and lays the groundwork for **federated**

directory and access infrastructure.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Requiring digital identity credentials from **trusted Credential Service Providers (CSPs)** such as Login.gov, ID.me, or CLEAR—aligned with **NIST 800-63-3 IAL2/AAL2 standards**—would significantly enhance both **cybersecurity and interoperability** across the healthcare landscape. These standards establish a consistent, high-assurance framework for identity proofing and authentication, creating a trusted foundation for health data exchange.

Cybersecurity Impacts:

1. **Reduced Risk of Identity Fraud and Unauthorized Access**
    - Verifying user identity at IAL2/AAL2 levels mitigates risks of impersonation and credential sharing—particularly important for accessing **sensitive clinical data and financial health information**.
2. **Improved Authentication Security Across Platforms**
    - Replacing weak or reused proprietary logins with **multi-factor, federated identity credentials** improves baseline security and aligns with **Zero Trust principles**.
3. **Enhanced Auditability and Incident Response**
    - CSP-issued credentials provide **robust audit trails** and revocation capabilities, allowing for more reliable detection and management of suspicious activity.

Data Exchange Impacts:

1. **Enables Scalable, Trusted Interoperability**
    - Federated digital identity supports **automated, role-based access control across networks, payers, providers, and third-party apps**,

reducing integration friction and manual authentication steps.

2. **Supports TEFCA Participation and Identity Assurance Requirements**

    - QHINs and other TEFCA entities must meet IAL2 requirements for Individual Access Services (IAS); CSP-based credentials allow **patients and providers to securely participate in national exchange**.

3. **Facilitates Consent and Delegated Access Workflows**

    - Trusted identities enable **accurate consent attribution**, support **caregiver or proxy access**, and enforce **patient-directed sharing of electronic health information** under 21st Century Cures Act requirements.

4. **Strengthens Trust in Digital Health Ecosystem**

    - Standardizing identity assurance builds **trust between data holders and requesters**, encouraging more open, secure participation in data exchange by payers, providers, and app developers.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

Mandating the use of the **OpenID Connect (OIDC)** identity protocol would provide a secure, standardized framework for authentication and identity federation across healthcare systems. It would simplify integration for payers, providers, and app developers by offering a **widely adopted, interoperable mechanism** to verify users—whether patients, caregivers, or clinicians—using trusted Credential Service Providers (CSPs). OIDC also supports **scalable, token-based authorization**, enabling smoother access to FHIR APIs and consent-managed workflows. Overall, its use would enhance **security, usability, and developer adoption**, helping unify identity management across platforms and support compliance with CMS and TEFCA requirements.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

CMS can encourage use of open, standards-based APIs by tying them to reimbursement, quality programs, and certification requirements (e.g., MIPS, MA Stars, Medicaid). Requiring FHIR-based APIs for data exchange, offering financial or regulatory incentives, and publishing conformance scores for vendors and payers would drive adoption.

Additionally, CMS can support developer toolkits, testing sandboxes, and clearer implementation guides to lower technical barriers.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider directory of **FHIR endpoints linked to verified digital identities** would significantly improve **data routing, care coordination, and patient access** by enabling automated, trusted connections across payers, providers, and apps.

It would reduce errors, streamline referrals and prior auth workflows, and enhance TEFCA and API-based exchange. **ONC or CMS** should maintain the directory to ensure trust and neutrality. Access should be **free or low-cost** for public benefit, especially for small providers, payers, and developers working to improve health equity and interoperability.

TD-6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

TEFCA uniquely enables **nationwide, standardized data exchange** through a common trust framework and **QHIN-to-QHIN connectivity**, supporting broad, secure interoperability. Its future support for **Individual Access Services** also distinguishes it from other networks.

a. Alternatives:
Existing options include **Carequality, CommonWell, DirectTrust, regional HIEs**, and **FHIR APIs** mandated by CMS. While effective in certain contexts, these are often limited by **scope, standardization, or vendor alignment**, making TEFCA critical for nationwide interoperability.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

Yes—several **redundant standards and exchange channels** could be consolidated to reduce complexity and improve interoperability. For example, **C-CDA and FHIR** are both used for clinical data exchange, but FHIR is more scalable and suited for modern APIs.

Similarly, **Direct Secure Messaging**, regional HIE protocols, and emerging **TEFCA frameworks** often overlap. Consolidating around **FHIR-based APIs and TEFCA-aligned governance** would streamline development, reduce integration burden, and support more efficient, real-time data access.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

a. Does it contain the full extent of data elements you need?

USCDI has meaningfully improved interoperability by **standardizing core data elements** for health information exchange, enabling more consistent FHIR API development and regulatory alignment. It has supported better data sharing across providers, payers, and third-party apps—especially for clinical summaries, labs, and medications.

a. Limitations and Missing Elements
While USCDI provides a strong foundation, it **does not yet include all the data elements needed** for comprehensive care coordination, value-based payment, or SDOH integration. Key gaps include:

- **Prior authorization and coverage details**
- **Behavioral health and substance use data (with consent)**
- **Detailed SDOH factors and interventions**
- **Device data, remote monitoring, and PGHD**
- **Care team roles and communication preferences**

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

The current limitations stem from **both the scope of the USCDI definition and how it is applied in practice**. USCDI prioritizes general clinical data but often excludes **payer, administrative, and real-world data** needed for full interoperability across the healthcare ecosystem. Additionally, **slow implementation cycles and lack of**

alignment with evolving standards (e.g., FHIR maturity levels) limit its practical value in dynamic use cases like prior authorization or digital quality measurement.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Expanding USCDI would **add significant value** by enabling more robust and cross-functional data exchange, especially for **social needs, behavioral health, payer transactions, and care coordination**. However, it could create **scoping and implementation challenges** if not carefully governed.

To address these challenges:

- Use a **tiered or modular approach** (e.g., required vs. optional elements by use case).
- Align with existing **HL7 implementation guides** (e.g., Gravity, Da Vinci, CARIN).
- Ensure **stakeholder-driven prioritization** and give vendors **realistic timelines** for implementation.

HealthLX supports a flexible expansion model that balances innovation with feasibility to accelerate meaningful, standards-based interoperability.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

With recent advances in language models and AI-driven data extraction, there is growing openness to **less structured but non-proprietary formats**—especially if it means capturing **richer, more complete clinical narratives**. While structured data (e.g., FHIR/USCDI) remains essential for automation and interoperability, flexible formats like **free text with standardized metadata** can enhance **contextual understanding, patient voice, and complex care documentation**.

That said, this approach would shift more burden to data receivers, so CMS should support **guidance, tooling, and validation frameworks** to ensure quality, security, and consistent interpretation. HealthLX supports a **hybrid model**, where structured fields are prioritized but **unstructured inputs are leveraged intelligently using NLP and AI**.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

The ONC Health IT Certification Program has significantly advanced interoperability and patient access through several key criteria and standards:

- **Standardized APIs for Patient and Population Services**: Mandating FHIR-based APIs, as per 45 CFR §170.315(g)(10), has enabled secure, real-time data exchange between EHRs and third-party applications, enhancing patient engagement and care coordination.UL Solutions+2McDermott+2eCFR+2

- **United States Core Data for Interoperability (USCDI) v3**: Serving as the baseline for data classes and elements, USCDI v3 promotes consistent data sharing across systems, facilitating comprehensive patient records and supporting public health initiatives.

- **Privacy and Security Criteria**: Requirements outlined in §170.315(d) ensure that certified health IT systems maintain robust protections for electronic health information, aligning with HIPAA standards and bolstering patient trust.Ropes & Gray+3eCFR+3McDermott+3

- **Insights Condition and Maintenance of Certification**: This condition mandates that certified health IT developers report on interoperability metrics, providing transparency and fostering continuous improvement in health IT performance.Ropes & Gray+2McDermott+2ND interactive+2

Collectively, these criteria and standards have been instrumental in promoting a more interoperable, secure, and patient-centered healthcare ecosystem.

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Redefining certification to focus on **API-enabled capabilities** rather than traditional software functionality would significantly enhance **modularity, innovation, and interoperability**. APIs allow health IT systems to **plug into broader digital ecosystems**, making them more adaptable and extensible across diverse care settings.

This approach reduces vendor lock-in, encourages **third-party app development**, and supports **real-time data exchange**—particularly important for FHIR-based workflows, patient access, and care coordination. It also ensures that certification aligns with **modern, cloud-based architectures**, enabling health IT to evolve rapidly with emerging technologies and patient expectations.

b. What would be the drawbacks?

While API-first certification promotes interoperability, it may introduce **new burdens and risks** for vendors and implementers. Smaller organizations could face **resource challenges** adapting to complex security, authentication, and maintenance requirements for APIs. Without strong enforcement, some APIs may be technically compliant but **functionally limited or poorly supported**, reducing real-world utility.

Additionally, a narrow API focus might **overlook essential in-system functionality** (e.g., clinical decision support, usability, offline access) that still matters in day-to-day care delivery. A balanced approach is needed—prioritizing APIs while maintaining baseline standards for software quality and performance.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

ASTP/ONC can revise certification criteria by requiring that certified APIs support **comprehensive access to all EHI**, including **structured, semi-structured, and unstructured data**. This includes:

- Mandating FHIR APIs expose not only discrete data (e.g., labs, meds), but also **free text notes, scanned documents, and imaging metadata**, using resources like DocumentReference, Binary, and Observation.

- Requiring health IT systems to implement **natural language processing (NLP) compatibility** or structured tagging for free-text content to make it retrievable and indexable via API.

- Ensuring certification includes **real-world testing for data completeness**, not just format compliance—verifying that all parts of a patient’s chart are accessible

through APIs under the **information blocking definition of EHI**.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

CMS can motivate providers to respond to API-based data requests with greater completeness and quality by **aligning data sharing with existing incentives and accountability mechanisms**. Key policy levers include:

- **Tie API response quality to value-based care programs**, such as MIPS, ACO REACH, and MA Stars, by rewarding providers who share timely, complete, and standardized data.

- **Integrate API responsiveness into quality reporting and certification metrics**, using real-world API usage audits to track provider performance.

- **Offer reimbursement or bonus payments for providers who expose full EHI** (including notes, scanned records, and imaging summaries) via certified APIs.

- Require participation in **national FHIR-based infrastructure (e.g., TEFCA, QHINs)** as a condition for favorable program participation.

HealthLX supports these goals by enabling providers to meet API data demands through FHIR-native orchestration, normalization, and consent-aware exchange.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

EHRs with **FHIR Bulk Data ($export) capabilities** can significantly reduce provider burden by enabling **centralized, automated quality measurement**—allowing CMS or intermediaries to extract population-level clinical data directly for quality performance analysis, rather than relying on manual provider submissions.

Capabilities Needed:

- Support for **FHIR R4 $export operations** using Group resources for attributed populations.

- Inclusion of all required quality data elements, including **lab results, medications, problem lists, and encounter details**.

- **Metadata on provenance and completeness** to ensure data integrity for CMS-calculated metrics.

Key Benefits:

- Reduces administrative overhead for providers.

- Improves consistency and accuracy of reported data.

- Enables **real-time performance feedback loops** if paired with dashboards or provider-facing apps.

Concerns:

- Not all EHRs fully support bulk export, especially in smaller or rural settings.

- Requires **robust patient attribution logic** and consistent data mapping across systems.

- Raises privacy and security considerations when **large-scale data exports** are moved across networks.

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

To fully implement the Cures Act requirement that certified APIs allow **access, exchange, and use of all EHI without special effort**, ASTP/ONC should take the following steps:

1. **Mandate Full EHI Exposure via FHIR APIs**

    - Extend certification to require that APIs **support all EHI elements**, including **free text notes, scanned documents, and imaging metadata**,

using FHIR resources like DocumentReference and Binary.

2. **Require Real-World Testing and Transparency**

   - Implement **real-world conformance testing** for API completeness, performance, and usability—beyond technical capability checks.

   - Publish **API performance scorecards** to promote transparency and accountability.

3. **Standardize Authentication and Access Models**

   - Require support for **SMART on FHIR, OAuth 2.0, and OpenID Connect** to simplify patient and third-party app access without proprietary barriers.

4. **Prohibit Restrictive Practices**

   - Enforce policies that prevent **fees, throttling, licensing terms, or custom integrations** that make accessing data via APIs more difficult than necessary.
   - Encourage EHR vendors to better support incorporation of outside data sources to augment the patient record. Most EHR vendors establish a black box model where only data created by their system can easily be seen by users. Outside sources are often relegated to isolated documents that make it difficult for users to gain valuable patient insights.

5. **Align API certification timelines with USCDI expansion**

   - Ensure developers **update APIs promptly as USCDI evolves**, so newly required data elements are made accessible without delay.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or “EHI export” for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

Yes—CMS and ASTP/ONC should revise the EHI export capability to require **standardized, FHIR-based API implementation**. Leaving the format and mechanism to individual health IT developers has led to **inconsistencies, limited automation, and added integration complexity** for providers, payers, and patients. Standardizing EHI export using **FHIR Bulk Data APIs ($export)** and **individual patient access APIs** would improve interoperability, ensure **data usability across platforms**, and reduce the burden on downstream users. This alignment would also reinforce compliance with the Cures Act's intent to enable **access and exchange without special effort**.

b. Are there specific workflow aspects that could be improved?

Yes—several workflow aspects of the current EHI export capability could be improved to reduce burden and improve usability:

- **Automation and Scheduling:** Allow for **automated, recurring exports** (rather than manual, ad-hoc requests) to better support quality reporting, payer access, and patient-directed exchange.

- **Status Tracking and Notifications:** Include **real-time export status updates and completion alerts** so users are not left waiting without confirmation.

- **Granular Export Controls:** Enable users to define **timeframes, data types, or patient subsets** for targeted exports, avoiding excessive or irrelevant data loads.

- **Consistent Export Format Descriptions:** Require developers to publish **machine-readable export specifications** (e.g., OpenAPI or FHIR CapabilityStatements) to support easier integration by external systems.

c. Should CMS consider policy changes to support this capability's use?

Yes—CMS should consider policy changes to **encourage routine, effective use of the EHI export capability**. This could include:

- **Incentivizing adoption** through quality programs (e.g., ACO REACH, MIPS) that reward providers for enabling **timely, standards-based data sharing**.

- **Aligning export capability with payer interoperability rules**, allowing exports to fulfill reporting, risk adjustment, and continuity of care use cases.

- **Requiring patient-facing tools** to leverage certified exports, especially in support of **data portability and care transitions**.

- **Promoting audit and compliance use** by encouraging providers to use exports for **internal quality review and patient record fulfillment** under HIPAA.

CMS can also publish **model workflows and use cases** to encourage consistent, real-world implementation across vendors and care settings.

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

Yes—CMS should endorse **non-CMS data sources and networks** (e.g., HIEs, QHINs, payer networks, and trusted app platforms) to accelerate **comprehensive, patient-centered data exchange**. Endorsement would help **standardize trust**, encourage broader participation, and improve care continuity across settings.

Recommended Criteria and Metrics for Endorsement:

- **Standards Compliance:** Participation in **HL7 FHIR®, USCDI, TEFCA, and ONC certification programs**.

- **Data Quality and Completeness:** Proven ability to exchange **accurate, timely, and comprehensive health data**, including structured and unstructured content.

- **Interoperability Performance:** Support for **cross-network exchange, Bulk FHIR exports, and SMART on FHIR apps**.

- **Patient Access and Consent Management:** Capability to support **individual access services, proxy workflows, and granular consent**.

- **Security and Identity Assurance:** Use of **NIST 800-63-3 IAL2/AAL2 digital identity standards** and robust authentication frameworks.

HealthLX supports this approach by acting as a neutral, standards-based platform that connects CMS-regulated systems with high-performing external networks and developers.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

APIs that provide access to the entirety of a patient's electronic health information (EHI) would unlock transformative opportunities across healthcare, including:

- **Comprehensive care coordination** by enabling all providers to see the full picture—structured and unstructured—at the point of care.
- **Improved patient empowerment and navigation**, giving beneficiaries and caregivers seamless access to records across settings.
- **Streamlined payer workflows**, supporting risk adjustment, prior auth, and value-based care with complete clinical insight.
- **Advanced analytics and AI**, powered by full data sets to improve population health, equity, and personalization.

a. What are the primary obstacles to this?

1. **EHR and Vendor Limitations:** Many systems still lack **complete FHIR API support** or restrict access to free text, scanned docs, and imaging.
2. **Lack of Standardization for All Data Types:** Not all EHI elements are **mapped to mature FHIR resources**, especially unstructured content.
3. **Privacy, Consent, and Legal Complexity:** Accessing the full EHI raises **heightened privacy risks**, especially for behavioral health and sensitive notes.
4. **Data Quality and Usability Variability:** Incomplete or inconsistent documentation limits **clinical trust and application usability**.

5. **Technical Burden on Small Providers:** Implementing full EHI APIs with **security, audit, and access controls** can be cost-prohibitive.

HealthLX's Role:
HealthLX bridges these gaps through a **FHIR-native, consent-aware orchestration platform**, helping organizations expose and consume full EHI responsibly, securely, and at scale.

b. What are the primary tradeoffs between USCDI and full EHI, especially (printed page 21040) given more flexible data processing capabilities today?

The **primary tradeoff** between using USCDI and accessing full EHI lies in balancing **standardization and ease of processing** with **completeness and contextual richness**.

- **USCDI** offers a **structured, curated subset** of health data ideal for consistent exchange, regulatory compliance, and automation—but it may omit clinically relevant data (e.g., free text notes, scanned reports, historical data) critical for decision-making.

- **Full EHI** provides a **more comprehensive patient record**, enabling better insight, especially for complex cases or AI applications—but often includes **unstructured or inconsistently documented elements** that require advanced processing (e.g., NLP) and pose privacy and usability challenges.

Given today's improved data processing tools (e.g., large language models, FHIR-native NLP engines), **full EHI is increasingly usable**, but still requires **clear governance, context tagging, and security protocols** to be practical at scale. **HealthLX recommends a hybrid approach**—using USCDI as a baseline for structured exchange, while enabling **on-demand, consented access to full EHI** for advanced use cases.

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

HealthLX's platform connects to **hundreds of FHIR API endpoints** across a growing network of **payers, providers, EHR vendors, and intermediary platforms**. These include:

- **Types:** Payer API endpoints (Patient Access, Provider Access, Payer-to-Payer), provider EHR APIs (clinical and scheduling), and TEFCA-ready exchange hubs.

- **Categories:** Acute and post-acute facilities, primary care practices, specialists, ACOs, and Medicare Advantage plans.

- **Geography:** Nationwide reach, with concentration in **Medicare and Medicaid-participating organizations** across urban and rural regions.

Endpoints are **searchable and indexable** by **National Provider Identifier (NPI), organizational ID, and FHIR CapabilityStatement metadata**, allowing HealthLX to dynamically route queries, manage consent, and verify endpoint trust status.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

HealthLX establishes patient data sharing connections through a combination of **standards-based and supported proprietary methods**, including:

- **FHIR §170.315(g)(10) Certified Endpoints**: Connections with ONC-certified EHR and payer APIs using **OAuth 2.0 + SMART on FHIR** protocols for secure access to clinical and claims data.

- **Bulk FHIR ($export) APIs**: For population-level queries aligned with payer-to-provider and quality reporting use cases.

- **TEFCA-Aligned HIE or QHIN Integrations**: Using **IHE XCA (Cross-Community Access)** standards and Direct protocols for document-based exchange, with transition plans toward TEFCA FHIR support.

- **Proprietary APIs and Intermediary Connections**: In cases where FHIR is not fully supported, HealthLX integrates through **vendor-specific APIs** and normalizes the data to FHIR internally for downstream use.

This multi-modal approach ensures **broad interoperability** while enabling a smooth transition toward full **FHIR-native exchange** as adoption matures.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

Yes—HealthLX actively **interconnects with other networks** to support nationwide, standards-based data exchange. These interconnections are established through:

- **Private Trust Agreements** with EHR vendors, payers, and HIEs for FHIR-based API access, consent management, and secure routing.
- **Participation in TEFCA-aligned Initiatives**, with readiness to integrate as QHINs come online, using TEFCA's **Common Agreement and QHIN-to-QHIN exchange protocols**.
- **Collaborations with Carequality and CommonWell**, where applicable, to support legacy C-CDA exchange and gradually transition toward FHIR-native workflows.
- **Vendor and Intermediary Partnerships** that provide bridge access to non-FHIR networks, which are then normalized to FHIR within the HealthLX platform.

This flexible approach enables **inter-network data sharing** across diverse technical environments while ensuring **governance, consent, and identity verification** are consistently enforced.

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

Increased adoption of **bulk FHIR APIs ($export)** would significantly enhance **population-level data access**, enabling more efficient and scalable support for key use cases such as:

- **Quality reporting and digital measurement** (e.g., eCQMs, HEDIS)
- **Risk adjustment and payment model analytics**

- **Population health and care management programs**

- **Public health surveillance and registry reporting**

Bulk FHIR allows structured data to be retrieved across attributed patient groups in **standardized, machine-readable formats**, reducing the need for custom extracts, manual reporting, or piecemeal queries. It also improves **timeliness, accuracy, and automation**, especially when paired with real-time feedback mechanisms and payer-provider collaboration tools.

b. What are the potential disadvantages of their use?

While bulk FHIR APIs offer powerful benefits, their use presents several challenges:

- **Technical Complexity and Infrastructure Load**: Bulk data queries can strain EHR systems not optimized for large-scale export, leading to **performance degradation** or delays.

- **Data Governance and Attribution Issues**: Accurately defining and managing **Group resources** for patient attribution requires coordination and may introduce **data consistency or ownership concerns**.

- **Privacy and Security Risks**: Exporting large datasets increases the stakes for **breach potential**, especially if access controls, encryption, or retention policies are weak.

- **Incomplete or Inconsistent Data**: Not all systems support **full FHIR resource maturity**, leading to variability in what's included across providers or vendors.

HealthLX mitigates these risks by providing a secure orchestration layer that validates, normalizes, and monitors bulk FHIR transactions across stakeholders.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

Maintaining **point-to-point models** offers direct control and customization for each connection, but it **does not scale efficiently**, requiring high operational overhead, redundant interfaces, and inconsistent standards enforcement. It also limits broad data liquidity and makes **cross-organizational coordination more difficult**.

In contrast, **shared network infrastructure** (e.g., TEFCA, QHINs, or trusted FHIR exchange hubs) enables **standardized governance, centralized trust, and scalable interoperability**. It simplifies onboarding, reduces duplication, and supports **nationwide patient access and data portability**—but may require **greater upfront alignment, shared policies, and reliance on third-party infrastructure**.

HealthLX supports a hybrid approach, enabling organizations to maintain local control while integrating into broader, standards-based networks for efficient, scalable exchange.

a. Do current rules encourage scalable network participation?

Current rules—including CMS's Interoperability and Patient Access Rule and ONC's Cures Act Final Rule—**lay important groundwork** by mandating FHIR APIs and discouraging information blocking. However, they **fall short of fully incentivizing scalable network participation**. Key limitations include:

- Lack of **enforceable participation requirements** in TEFCA or other trusted exchange frameworks.

- Minimal incentives or mandates for smaller providers, payers, or health IT vendors to adopt **shared infrastructure** over point-to-point approaches.

- No clear alignment between **compliance programs and real-world interoperability performance**, such as volume of successful data exchange or QHIN engagement.

To drive scalable participation, CMS could tie **network participation to value-based care programs**, publish **network performance metrics**, and support **shared services for smaller organizations**.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

To improve alignment and reduce fragmentation, CMS and ONC should pursue the following changes:

- **API Unification Across Programs**: Require consistent implementation of FHIR APIs (e.g., Patient Access, Provider Access, Payer-to-Payer) using shared standards and authentication models, minimizing duplication across regulatory

programs.

- **Mandate Reciprocal Access**: Enforce policies that require **any entity accessing data via APIs or networks to also make their data accessible** under the same conditions—supporting equity and reducing one-way data flow.

- **Align Program Incentives with Network Participation**: Tie participation in **TEFCA, QHINs, or certified FHIR exchange hubs** to eligibility for CMS quality and value-based care programs.

- **Support Federated Identity and Consent Models**: Promote adoption of **NIST-aligned digital identity and consent frameworks**, enabling seamless, role-based data sharing across networks.

- **Encourage Use of Shared Directory and Endpoint Services**: Standardize use of national **FHIR endpoint directories** to streamline routing and reduce point-to-point configuration burdens.

HealthLX's Role:
HealthLX supports these changes by providing a **scalable, API-first interoperability layer** that enables reciprocal, secure data exchange across both point-to-point and shared network models.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

CMS and ONC should play an active role in ensuring that healthcare data sharing networks are **financially viable, widely adopted, and outcome-driven**. Sustaining these networks requires balancing operational costs with **broad participation and demonstrable value**.

Key actions CMS and ONC should take include:

- **Stimulate Demand via Program Alignment**: Tie participation in trusted networks (e.g., TEFCA, QHINs) to **CMS reimbursement, quality programs, and regulatory compliance**, ensuring consistent demand.

- **Fund Core Infrastructure for Underserved Regions**: Provide **technical assistance or grants** to support rural and small providers in onboarding to shared networks.

- **Monitor Network Performance and Usage Metrics**: Require networks to report on **data exchange volume, response times, participant diversity**, and real-world impact on care coordination and patient access.

- **Foster Market Competition and Transparency**: Prevent data monopolies by enforcing **reciprocal access, standards adherence, and fair pricing**, creating a healthy supply of interoperable services.

- **Encourage Public-Private Collaboration**: Support innovation through **open APIs, shared directories, identity frameworks, and pilot programs** that reduce the burden of entry for new participants.

HealthLX supports this vision by operating as a neutral, FHIR-native interoperability platform that connects both emerging and established networks, enabling sustainable, standards-based data exchange at scale.

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

As a health IT interoperability vendor, HealthLX has encountered several practices that may constitute information blocking under 45 CFR Part 171. These include:

1. Non-Responsiveness or Delayed Access

- **Example:** EHR vendors or provider organizations failing to respond to FHIR API access requests for days or weeks, even after all legal requirements (e.g., patient authorization, security validation) were met.
- **Impact:** Hinders timely patient data access and disrupts third-party application workflows, especially in care transitions or time-sensitive interventions.

2. Partial or Incomplete API Implementation

- **Example:** FHIR APIs that are technically "available" per certification but **exclude key USCDI elements** or return only placeholder data (e.g., empty fields or generic metadata).
- **Impact:** Results in unusable or misleading data for clinical decision support, quality reporting, or payer risk scoring.

3. Arbitrary API Usage Restrictions or Throttling

- **Example:** Health IT developers requiring manual review and approval before each new API transaction, or limiting API call frequency **far below what is needed for population-level use**.
- **Impact:** Makes patient- or provider-facing applications functionally inoperable despite having access "in name only."

4. Denial of Patient-Directed Access via Apps

- **Example:** Providers refusing to enable patient app connections or requiring the use of **their own branded portal**, even when patients request access via a certified third-party SMART on FHIR app.
- **Impact:** Violates the principle of patient-directed exchange and creates unnecessary technical and legal barriers.

HealthLX supports proactive enforcement and clearer implementation guidance to ensure APIs and data access practices meet the intent of the 21st Century Cures Act.

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

To further discourage information blocking, ASTP/ONC and CMS should implement policies that **increase accountability, transparency, and alignment with reimbursement programs**. Key recommendations include:

1. Tie Data Sharing to Financial Incentives and Program Eligibility

- Make full participation in **CMS programs (e.g., MIPS, ACO REACH, MA Stars)** conditional on **demonstrated, standards-compliant data sharing**, not just attestation.

2. Publish Provider Compliance and Blocking Reports

- Create **public scorecards or dashboards** showing provider-level API performance, information blocking complaints, and participation in trusted networks (e.g., TEFCA, QHINs).

3. Strengthen Real-World Testing and Audits

- Conduct **randomized audits of API responses and patient-directed access workflows** to ensure providers meet both the letter and spirit of compliance.

4. Establish Escalating Corrective Action Frameworks

- Use a **graduated enforcement model** that begins with education and technical assistance but progresses to **fines or program exclusion** for repeat or egregious violations.

5. Clarify Proxy Access and App-Directed Sharing Rules

- Require providers to support **authorized caregiver and app-based data access** under clear guidance, reducing legal uncertainty often cited as a reason to deny access.

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

Yes—certain categories of **healthcare actors covered under PHSA §3022(a)(1)** lack meaningful disincentives for engaging in information blocking. Notably:

- **Post-acute and long-term care providers** (e.g., skilled nursing facilities, home health agencies) are **not yet subject to many CMS interoperability requirements**, yet they hold critical patient data.

- **Behavioral health providers and substance use treatment programs** often fall outside the reach of **ONC certification and CMS payment models**, despite being included in the statutory definition of a health care provider.

- **Small or unaffiliated physician practices** may be technically subject to the law but lack **targeted oversight or performance-based disincentives**.

- **Health IT developers not certified under the ONC Health IT Certification Program** may not face enforcement unless participating in federal programs, creating **uneven accountability**.

To promote fair compliance, CMS and ONC should **expand program-linked disincentives** and consider **graduated enforcement** for non-hospital providers that hold patient data but lack sufficient interoperability pressure.

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

Despite regulatory efforts, price transparency data remains **difficult to access, interpret, and use meaningfully** due to several persistent shortcomings:

- **Content Gaps**: Many hospitals and payers omit required items/services or provide **incomplete negotiated rate data**, especially for bundled services and shoppable procedures.

- **Inconsistent Formats**: Machine-readable files vary widely in structure, naming conventions, and terminology, lacking **standardized schemas** that allow developers to create usable tools.

- **Poor Delivery and Accessibility**: Pricing files are often buried deep on websites, require manual discovery, or are **not updated regularly**, making them inaccessible to patients and app developers.

- **Timeliness Issues**: Updates are inconsistent and **rarely reflect real-time costs**, especially for dynamically priced services, negotiated rates, or geographically adjusted fees.

- **Lack of Integration into Care Workflows**: Pricing data is not embedded in EHRs or patient portals in a way that enables **real-time cost estimation at the point of care or referral**.

HealthLX supports structured APIs and integration models that would allow pricing data to be dynamically retrieved and applied within care management, prior authorization, and patient-facing tools.

b. Which workflows would benefit most from functional price transparency?

Price transparency would have the greatest impact in workflows where cost directly affects care decisions—like **referrals, prior authorization, scheduling, and patient financial counseling**. Real-time, usable pricing data at these points would help patients choose affordable options, improve adherence, and reduce surprise billing. Embedding this data into **EHRs, portals, and digital tools** transforms cost from a barrier into an informed decision point.

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

The most valuable improvements would be:

- **Standardized formats** for machine-readable pricing files to ensure consistency across payers and providers;

- **API-based access** to real-time pricing data embedded in EHRs and patient tools;

- **Clear out-of-pocket estimates** tailored to individual coverage and service location;

- And **enforcement of timely updates and data completeness**.

These changes would empower patients to make informed choices, help providers guide cost-aware care, and enable CMS to monitor market behavior and promote value-based care more effectively.

d. What would further motivate solution development?

Solution development would accelerate if CMS offered **clear technical standards, enforcement mechanisms, and financial or regulatory incentives**. Developers need **stable, accessible, real-time pricing APIs**, aligned with patient and provider workflows. Additional motivators include **integration requirements tied to value-based care programs**, and **recognition or certification for compliant tools**, which would increase adoption and market trust.

F. Value-Based Care Organizations

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve value-based care organizations. While we certainly want value-based care organizations to answer questions in this section (and in other sections) from the value-based care provider point of view, we also invite all stakeholders to provide their viewpoints on the value-based care workflows as appropriate.

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

To accelerate adoption of digital health and care navigation tools in ACOs and MSSP models, CMS should align **incentives with VBC performance metrics** and reduce integration barriers.

Recommended Incentives:

- **Bonus points or shared savings credit** for ACOs that deploy certified digital tools to improve patient engagement, chronic care management, or reduce unnecessary utilization.

- **Reimbursement for digital interventions**, such as remote monitoring, app-based care plans, or AI-guided triage tools.

- **Inclusion of digital health adoption as a reportable metric** in MSSP dashboards or quality benchmarks.

Current Obstacles:

- **Fragmented EHR and API ecosystems**, making integration of third-party apps difficult and costly.

- **Limited access to real-time, complete patient data**, especially across care settings and systems.

- **Lack of funding or technical infrastructure** for small ACOs to pilot or scale digital solutions.

- **Patient engagement barriers**, including digital literacy, language access, and trust in virtual tools.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

CMS can better integrate AI, population health tools, and care coordination technologies into APMs by **embedding these capabilities directly into performance and reporting requirements**. Key actions include:

- **Incorporate digital capabilities into APM scoring models**, such as awarding credit for using AI-powered risk stratification, care gap closure tools, or predictive analytics for population health management.

- **Mandate FHIR-based APIs and data exchange standards** to support seamless integration of analytics platforms, remote monitoring, and patient engagement apps into ACO workflows.

- **Tie quality measurement and shared savings to demonstrable use of digital tools** that improve care coordination (e.g., closed-loop referrals, real-time alerts, care navigation support).

- **Encourage usability by requiring human-centered design certification** or evidence of equitable patient engagement across digital channels.

These steps would align innovation with measurable outcomes, reduce administrative burden, and help ACOs and other APM participants use technology to drive success in value-based models.

VB-3. What are essential health IT capabilities for value-based care arrangements?

Value-based care arrangements require **interoperable, real-time, and patient-centered** health IT capabilities. The most essential include:

- **FHIR-native data exchange** to support scalable, standards-based integration across payers, providers, and care teams.
- **Bulk data APIs** for population-level reporting, risk stratification, and performance measurement.
- **Real-time care coordination tools**, including closed-loop referral tracking, alerts, and shared care plans.
- **Patient and caregiver access** via digital identity, consent, and app-based navigation platforms.
- **Analytics and quality dashboards** that visualize gaps in care, cost trends, and outcomes by cohort.
- **Support for remote monitoring and SDOH capture**, to extend care beyond clinical settings and address equity goals.

These capabilities are foundational to improving outcomes, lowering cost, and driving performance in accountable care and shared savings models.

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

To succeed in value-based care arrangements, organizations must implement health IT solutions that support **timely, accurate, and actionable data exchange across the care continuum**. Key capabilities include:

- **Care Planning & Coordination**: Tools for creating, sharing, and updating patient care plans across multidisciplinary teams using FHIR-based APIs.

- **Patient Event Notification**: Real-time alerts for admissions, discharges, ED visits, and other critical transitions to reduce readmissions and close care gaps.

- **Data Extraction & Normalization**: Automated ingestion and standardization of data from EHRs, payers, HIEs, and remote sources into a unified, FHIR-native format.

- **Quality Performance Measurement**: Dashboards and analytics tools that enable automated tracking of metrics (eCQMs, HEDIS, Stars) across attributed populations.

- **Claims Data Access**: Integration with CMS APIs (e.g., Blue Button 2.0) and payer data feeds to inform risk scoring, utilization analysis, and cost containment strategies.

- **Attribution and Patient Matching**: Accurate identification and alignment of patients to providers or ACOs using trusted identifiers and referential matching logic.

- **Remote Device Interoperability**: Connectivity to wearables and home health devices for remote monitoring, chronic disease management, and timely intervention.

- **Patient Empowerment Tools**: SMART on FHIR apps, portals, and caregiver proxy access mechanisms to increase engagement, adherence, and shared decision-making.

HealthLX supports all of the above functions through it's composable automation interoperability platform, enabling scalable, consent-managed data exchange across clinical, payer, and patient systems.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

Beyond core interoperability, the following health IT capabilities have proven highly valuable in helping organizations thrive in value-based care models:

- **Consent Management and Proxy Access**: Systems that handle granular, role-based data access—including caregiver and patient-directed sharing—are essential for compliant, inclusive data exchange.

- **Identity Verification and Federated Login**: Use of NIST-aligned digital identity credentials (e.g., ID.me, Login.gov) ensures secure, scalable access to data across providers and digital health tools.

- **Integrated SDOH Data Capture and Exchange**: Tools that incorporate social needs assessments and referrals using standardized codes (e.g., Gravity Project FHIR IGs) to support equity-focused care.

- **Cross-Network Routing and Endpoint Discovery**: Capabilities that dynamically resolve FHIR endpoints and support routing across QHINs, HIEs, and EHR vendor networks.

- **Real-Time Prior Authorization Support**: Integration of CDS Hooks and Da Vinci-guided prior auth APIs that reduce care delays and administrative burden.

- **Clinical-Operational Integration**: Platforms that bridge care delivery and back-office workflows—such as aligning clinical data with billing, risk adjustment, and quality submission processes.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

Successful participation in value-based care arrangements relies on access to a comprehensive set of data types that support care coordination, risk adjustment, and quality performance. Essential data includes clinical information such as labs, vitals, diagnoses, procedures, care plans, and medications, as well as claims and encounter data to track utilization and cost. Demographics, insurance coverage, and quality measurement data (e.g., eCQMs, HEDIS, and Stars metrics) are also critical. Additional valuable inputs include behavioral health and substance use data (with appropriate consent), social determinants of health (SDOH) such as housing or food insecurity, patient-reported outcomes and goals, remote monitoring data, and care coordination status across referrals. Access to these data types in structured,

FHIR-compatible formats is key to enabling automation, reducing administrative burden, and achieving shared savings and outcome targets in value-based care models.

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

Current ONC Health IT Certification criteria have laid a strong foundation for value-based care by promoting standardized data exchange, API-enabled access, and baseline interoperability through USCDI and FHIR implementation. Certified capabilities like the standardized API for patient and population services and support for electronic clinical quality measures (eCQMs) help value-based care organizations meet reporting and data-sharing requirements. However, the certification program often focuses more on individual data access than on **population-level workflows** central to value-based models.

There is also limited emphasis on supporting capabilities such as **closed-loop referrals, social care integration, real-time alerts**, and bulk data exchange—all essential to care coordination and risk-based performance. To better serve VBC delivery, certification should evolve to more explicitly support **multi-stakeholder, team-based, and longitudinal care** use cases across diverse data sources and care settings.

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

While the ONC Health IT Certification Program has advanced baseline interoperability, several essential capabilities that directly support alternative payment models (APMs) and value-based care remain unaddressed. To enable more effective, scalable, and equitable care delivery, ONC should expand certification criteria to include the following:

1. Bulk FHIR ($export) for Population-Level Workflows

Certification should include standardized support for FHIR-based bulk data extraction across attributed populations. This is critical for risk stratification, digital quality reporting, and cost/utilization analysis—all foundational to APM performance.

2. Closed-Loop Referral and Care Coordination Tracking

Currently, there is no certification requirement for tracking referral status, follow-up, or shared care plan execution. These capabilities are essential for care continuity and reducing avoidable costs in accountable care arrangements.

3. Social Determinants of Health (SDOH) Integration

APMs increasingly require the ability to assess, document, and act on SDOH needs. Certification should support exchange of SDOH data using standards like the Gravity Project's FHIR implementation guides and enable connectivity with community-based service platforms.

4. Consent Management and Proxy Access Support

Effective patient and caregiver engagement requires certified tools for managing digital consent, role-based access, and proxy permissions—capabilities currently lacking in the certification program despite their importance for compliant, inclusive data exchange.

5. Real-Time Data Availability and Routing Infrastructure

Certification should include real-time API response performance, endpoint discovery, and routing capabilities across networks and vendors. Timeliness is crucial in APMs for coordinating interventions, avoiding duplicative care, and improving outcomes.

Conclusion:
Modernizing ONC certification to include these capabilities would directly enhance the utility of certified health IT in value-based programs, reduce friction in multi-organization data exchange, and accelerate digital innovation aligned with APM goals.

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

Technology requirements for APMs can reduce complexity and improve adoption by aligning around **core, standardized capabilities**—such as FHIR APIs, USCDI-aligned data elements, and digital identity protocols—while allowing flexibility in how those capabilities are implemented and extended.

CMS and ONC should define a **foundational set of interoperable functions** (e.g., patient access, bulk data export, quality measure reporting, care coordination support) that all participants must meet, whether through CEHRT or other modular, standards-based tools. By ensuring these requirements are **use-case driven and outcome-oriented**, rather than tied to rigid software certification alone, APMs can adapt technologies to their scale, setting, and patient population.

At the same time, **optional advanced capabilities**—such as closed-loop referrals, SDOH exchange, or predictive analytics—could be incentivized through program bonuses or performance scoring rather than mandated. This tiered approach ensures a **baseline of interoperability and data access**, while preserving flexibility for APMs to innovate with the tools that best fit their operational models.

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

HHS policies—beyond CEHRT—can play a pivotal role in closing interoperability gaps and optimizing the use of digital health products in APMs. For example, expanding existing **Conditions of Participation (CoPs)** requirements, such as the mandate to send electronic patient event notifications under 42 CFR 482.24(d), to **explicitly include APM-participating providers** (e.g., ACOs, community specialists, post-acute partners) would help standardize access to real-time data needed for timely interventions and care coordination.
Currently, APM participants often face barriers such as:

- **Lack of hospital participation in health information exchanges (HIEs)** or APIs that route event notifications beyond primary treating providers.

- **Variability in how EHR vendors implement and expose event notification capabilities**, creating inconsistencies across care settings.

- **Unclear requirements for non-hospital actors** (e.g., SNFs, FQHCs, specialists) to generate or receive such notifications under CMS programs.

- **Absence of aligned incentives or enforcement** to ensure delivery of actionable notifications to all members of the care team, particularly those outside the admitting facility's system.

HHS can address these barriers by:

- Requiring **bidirectional event notification participation** from hospitals and APM entities through updated CoPs or APM participation agreements.

- Supporting **standardized routing infrastructure** (e.g., QHINs, TEFCA frameworks) to deliver notifications to APM participants regardless of system affiliation.

- Offering **incentives for event notification readiness** through MSSP, ACO REACH, or Medicaid waiver programs.

These policy adjustments would ensure APM participants have the **real-time situational awareness** necessary to reduce readmissions, close care gaps, and improve outcomes—key goals of value-based care.

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

APM organizations operate under a performance-based model that requires **proactive care coordination, population health management, and outcomes-based accountability**—so their technology requirements must go beyond those of traditional, fee-for-service (FFS) entities.

Specifically, APM organizations should be required to support:

- **Bulk FHIR ($export) capabilities** for population-level reporting, digital quality measurement, and risk adjustment—critical for managing attributed panels and contract performance.

- **Real-time, multi-source data exchange** across care settings, including the ability to consume and respond to **event notifications, closed-loop referrals, and care plan updates**.

- **Integration of claims, clinical, and SDOH data** to support holistic risk stratification and care gap closure, using **standards such as USCDI+ and Gravity Project FHIR IGs**.

- **Patient engagement and caregiver support tools**, including digital identity, proxy access, and app-enabled self-management—beyond minimum patient portal requirements.

In contrast, non-APM organizations may only need to meet **point-in-time interoperability requirements** (e.g., patient access, clinical summaries) without demonstrating population-level or performance-driven capabilities.

To reflect these differences, CEHRT or other program criteria should be **tiered**, with **advanced interoperability, analytics, and engagement capabilities required for APMs**, while still allowing modular adoption pathways that fit organizational size and readiness.

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

CMS's approach to aligning CEHRT requirements with the clinical and operational realities of Advanced APMs is well-founded. To make this flexibility impactful while preserving data quality and care outcomes, CMS should define a **core set of CEHRT criteria applicable to all APMs**, and allow **tailored additions based on specialty or setting**.

Recommended Core Certification Criteria for All Advanced APMs:

- **FHIR API capabilities** (including Patient Access, Provider Access, and Bulk $export) to support real-time and population-level exchange.

- **Clinical quality measure reporting APIs** to automate submission of eCQMs, HEDIS, and APM-specific metrics.

- **Care coordination and closed-loop referral tracking** via certified use of Task, ServiceRequest, and CarePlan FHIR resources.

- **Support for digital identity and proxy access** for patient and caregiver engagement, especially in chronic or home-based care models.

Tailored Certification by Clinical Focus or Setting:

- For **primary care ACOs**: SDOH integration (Gravity FHIR IG), longitudinal care plans, risk stratification tools.

- For **behavioral health APMs**: Consent management APIs, privacy-sensitive data segmentation, and integration of behavioral and physical health records.

- For **post-acute or home health models**: Remote monitoring interoperability, ADT notification ingestion, and mobile-compatible engagement tools.

Flexibilities for Rural or Small APM Participants:

- Allow **modular certification pathways** that prioritize core data-sharing capabilities without requiring full-featured EHRs.

- Offer **shared technology infrastructure or regional hubs** (e.g., QHIN-based services or HIE intermediaries) to lower the burden of compliance.

- Provide **technical assistance funding or CMS-endorsed solutions** that meet baseline CEHRT criteria affordably and without high integration costs.

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

Implementing value-based care programs consistently reveals critical interoperability gaps that limit data access, slow performance measurement, and impede care coordination across settings. Specific challenges include:

- Fragmented data sources and inconsistent FHIR support across EHRs and payer platforms, requiring custom integrations and data normalization just to achieve baseline connectivity.
- Lack of real-time data availability, particularly for events like hospital admissions, discharges, or emergency visits—hindering timely intervention and transitional care management.
- Inadequate population-level data access, especially where Bulk FHIR APIs are unavailable or underperforming, making it difficult to automate quality reporting and risk adjustment workflows.
- Siloed behavioral health, SDOH, and community care data, which are essential for equity-focused care but often excluded from interoperable exchanges due to technical, regulatory, or policy barriers.
- Weak patient identity resolution across systems, complicating attribution, consent enforcement, and data deduplication for shared-risk models.
- Variable data completeness and provenance, which undermines trust in exchanged information and makes it difficult to act on care gaps or performance analytics with confidence.

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

The key to promoting innovation while preserving flexibility across APMs is to **standardize foundational technologies and data exchange protocols**, while allowing programs to tailor their use based on clinical context and population needs.

CMS and ONC should mandate the use of **core, open standards**, such as:

- **FHIR APIs** for clinical, administrative, and patient access workflows

- **USCDI and USCDI+** for consistent data element exchange across programs

- **Bulk FHIR ($export)** for population health and quality performance measurement

- **SMART on FHIR** for plug-and-play app integration

- **OAuth 2.0 and OpenID Connect** for secure, scalable authentication

These standards provide a **shared technical foundation** while still allowing APMs to innovate with specialty-specific tools, advanced analytics, digital engagement platforms, and modular reporting systems. In short, standardization should ensure **data liquidity and interoperability**, while program-level flexibility should define **how data is used to drive outcomes**.

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

To better support value-based care and reduce provider burden, CMS and ONC should focus on **streamlining data access, automating performance workflows, and aligning standards with real-world use cases**. Key improvements include:

- **Enhance support for Bulk FHIR ($export)** and require real-world testing to ensure it enables automated quality reporting, population health analytics, and risk adjustment—removing the need for manual data extraction.

- **Expand certification to include closed-loop care coordination capabilities**, such as referral tracking and care plan reconciliation, which are essential to ACOs and other APMs.

- **Standardize integration of claims and clinical data** via FHIR-based APIs, enabling more complete views of patient care and reducing reconciliation work between systems.

- **Include patient consent and proxy access capabilities in certification**, using SMART on FHIR and OAuth standards to reduce manual processing and improve data sharing with caregivers.

- **Simplify quality reporting through digital-first measures**, leveraging APIs that allow passive, automated reporting tied directly to clinical workflows—minimizing redundant documentation.

By shifting certification criteria toward **outcomes-driven, automation-friendly standards**, CMS and ONC can support innovation while alleviating the technical and administrative overhead often borne by providers in VBC arrangements.

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

Implementing secure, NIST-aligned digital identity credentials (e.g., ID.me, Login.gov, or other IAL2-compliant solutions) would significantly enhance the effectiveness of value-based care by enabling **trusted, seamless access to patient data across organizations**. This supports better care coordination, real-time decision-making, and patient engagement—all critical to achieving VBC outcomes.

Digital credentials reduce friction in data exchange by **ensuring patient, caregiver, and provider identities are consistently verified** across networks. This allows APMs and ACOs to confidently share sensitive health information, manage proxy access, and ensure compliance with privacy and consent regulations.
In addition, digital identity frameworks support **scalable interoperability**, facilitating connections across payers, providers, and third-party tools—without duplicative logins or manual verification. For patients, it enables easier access to care management platforms, engagement tools, and health records, improving adherence and satisfaction.

By embedding trusted identity verification into the VBC infrastructure, CMS can drive **greater data liquidity, reduce administrative burden, and accelerate equity-focused, patient-centered outcomes.**

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR (printed page 21041) endpoints directory to maximize its effectiveness?

A nationwide provider directory of FHIR endpoints would dramatically improve **data routing, interoperability, and access to both clinical and claims data**—especially for organizations operating under value-based care (VBC) models. By enabling automated discovery of where and how to connect to other systems, this directory

would reduce manual overhead, eliminate duplicate data requests, and **accelerate care coordination and quality measurement**.

For ACOs and APM participants, this capability would streamline connections to provider organizations, payers, and networks for **retrieving patient data, exchanging referrals, and analyzing utilization and cost** across the continuum of care.

Key Data Elements Required for Directory Effectiveness:

- **FHIR Endpoint URLs** (clearly indicating patient, provider, payer, and bulk data access)
- **Organization name, NPI, and location**
- **Endpoint purpose and capability statement** (e.g., supports US Core, SMART on FHIR, Bulk $export)
- **Affiliated networks or QHIN participation**
- **Digital certificate or trust framework details** (for secure connection validation)
- **Contact information for technical coordination**
- **Last validated date/status** to ensure freshness and reliability

A standardized, openly accessible directory—maintained or endorsed by CMS or ONC—would reduce friction in health data exchange and enable **faster, more accurate connections** that directly support VBC goals such as closing care gaps, tracking outcomes, and minimizing redundant services.

III. Collection of Information Requirements

Please note, this is a request for information (RFI) only. In accordance with the implementing regulations of the Paperwork Reduction Act of 1995 (PRA), specifically 5 CFR 1320.3(h)(4), this general solicitation is exempt from the PRA. Facts or opinions submitted in response to general solicitations of comments from the public, published in the Federal Register or other publications, regardless of the form or format thereof, provided that no person is required to supply specific information pertaining to the commenter, other than that necessary for

self-identification, as a condition of the agency's full consideration, are not generally considered information collections and therefore not subject to the PRA.

This RFI is issued solely for information and planning purposes; it does not constitute a Request for Proposal (RFP), applications, proposal abstracts, or quotations. This RFI does not commit the U.S. Government to contract for any supplies or services or make a grant award. Further, CMS and ASTP/ONC are not seeking proposals through this RFI and will not accept unsolicited proposals. Responders are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. CMS and ASTP/ONC note that not responding to this RFI does not preclude participation in any future procurement, if conducted. It is the responsibility of the potential responders to monitor this RFI announcement for additional information pertaining to this request. In addition, CMS and ASTP/ONC note that we will not respond to questions about potential policy issues raised in this RFI.

CMS and ASTP/ONC will actively consider input as we develop future regulatory proposals or future subregulatory policy guidance. We may or may not choose to contact individual responders. Such communications would be for the sole purpose of clarifying statements in the responders' written responses. Contractor support personnel may be used to review responses to this RFI. Responses to this notice are not offers and cannot be accepted by the Government to form a binding contract or issue a grant. Information obtained as a result of this RFI may be used by the Government for program planning on a non-attribution basis. Respondents should not include any information that might be considered proprietary or confidential. This RFI should not be construed as a commitment or authorization to incur cost for which reimbursement would be required or sought. All submissions become U.S. Government property and will not be returned. In addition, we may publicly post the public comments received or a summary of those public comments.

Stephanie Carlton, Deputy Administrator of the Centers for Medicare & Medicaid Services, approved this document on May 9, 2025.

Steven Posnack, Acting Assistant Secretary for Technology Policy, Acting National Coordinator for Health Information Technology, approved this document on May 6, 2025.

Robert F. Kennedy, Jr.,

Secretary, Department of Health and Human Services.